

In calculus we use radians to measure angles except when otherwise indicated. The following table gives the correspondence between degree and radian measures of some common angles.

Degrees	0°	30°	45°	60°	90°	120°	135°	150°	180°	270°	360°
Radians	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$	$\frac{2\pi}{3}$	$\frac{3\pi}{4}$	$\frac{5\pi}{6}$	π	$\frac{3\pi}{2}$	2π

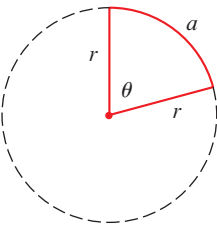


FIGURE 1

Figure 1 shows a sector of a circle with central angle θ and radius r subtending an arc with length a . Since the length of the arc is proportional to the size of the angle, and since the entire circle has circumference $2\pi r$ and central angle 2π , we have

$$\frac{\theta}{2\pi} = \frac{a}{2\pi r}$$

Solving this equation for θ and for a , we obtain

(3)

$$\theta = \frac{a}{r}$$

$$a = r\theta$$

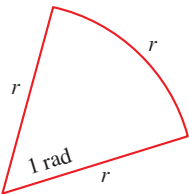


FIGURE 2

Remember that Equations 3 are valid only when θ is measured in radians.

In particular, putting $a = r$ in Equation 3, we see that an angle of 1 rad is the angle subtended at the center of a circle by an arc equal in length to the radius of the circle (see Figure 2).

EXAMPLE 2

- (a) If the radius of a circle is 5 cm, what angle is subtended by an arc of 6 cm?
- (b) If a circle has radius 3 cm, what is the length of an arc subtended by a central angle of $3\pi/8$ rad?

SOLUTION

- (a) Using Equation 3 with $a = 6$ and $r = 5$, we see that the angle is

$$\theta = \frac{6}{5} = 1.2 \text{ rad}$$

- (b) With $r = 3$ cm and $\theta = 3\pi/8$ rad, the arc length is

$$a = r\theta = 3\left(\frac{3\pi}{8}\right) = \frac{9\pi}{8} \text{ cm}$$

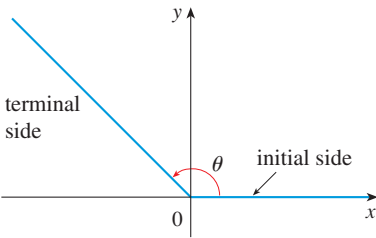


FIGURE 3 $\theta \geq 0$

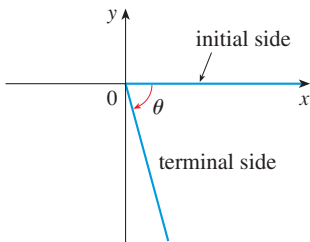


FIGURE 4 $\theta < 0$

The **standard position** of an angle occurs when we place its vertex at the origin of a coordinate system and its initial side on the positive x -axis as in Figure 3. A **positive** angle is obtained by rotating the initial side counterclockwise until it coincides with the terminal side. Likewise, **negative** angles are obtained by clockwise rotation as in Figure 4.

Figure 5 shows several examples of angles in standard position. Notice that different angles can have the same terminal side. For instance, the angles $3\pi/4$, $-5\pi/4$, and

$11\pi/4$ have the same initial and terminal sides because

$$\frac{3\pi}{4} - 2\pi = -\frac{5\pi}{4} \quad \frac{3\pi}{4} + 2\pi = \frac{11\pi}{4}$$

and 2π rad represents a complete revolution.

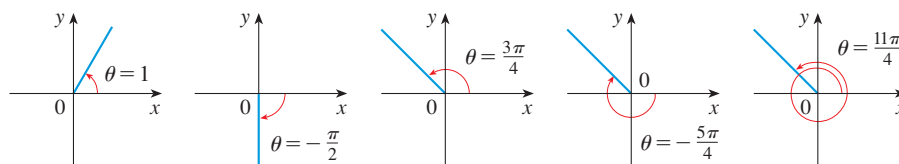


FIGURE 5
Angles in standard position

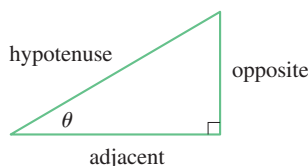


FIGURE 6

■ The Trigonometric Functions

For an acute angle θ the six trigonometric functions are defined as ratios of lengths of sides of a right triangle as follows (see Figure 6).

(4)

$$\begin{aligned} \sin \theta &= \frac{\text{opp}}{\text{hyp}} & \csc \theta &= \frac{\text{hyp}}{\text{opp}} \\ \cos \theta &= \frac{\text{adj}}{\text{hyp}} & \sec \theta &= \frac{\text{hyp}}{\text{adj}} \\ \tan \theta &= \frac{\text{opp}}{\text{adj}} & \cot \theta &= \frac{\text{adj}}{\text{opp}} \end{aligned}$$

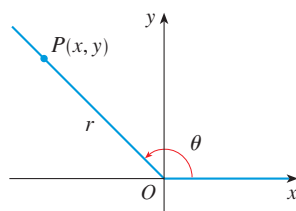


FIGURE 7

This definition doesn't apply to obtuse or negative angles, so for a general angle θ in standard position we let $P(x, y)$ be any point on the terminal side of θ and we let r be the distance $|OP|$ as in Figure 7. Then we define

(5)

$$\begin{aligned} \sin \theta &= \frac{y}{r} & \csc \theta &= \frac{r}{y} \\ \cos \theta &= \frac{x}{r} & \sec \theta &= \frac{r}{x} \\ \tan \theta &= \frac{y}{x} & \cot \theta &= \frac{x}{y} \end{aligned}$$

Since division by 0 is not defined, $\tan \theta$ and $\sec \theta$ are undefined when $x = 0$ and $\csc \theta$ and $\cot \theta$ are undefined when $y = 0$. Notice that the definitions in (4) and (5) are consistent when θ is an acute angle.

If θ is a number, the convention is that $\sin \theta$ means the sine of the angle whose *radian* measure is θ . For example, the expression $\sin 3$ implies that we are dealing with an angle of 3 rad. When finding a calculator approximation to this number, we must remember to set our calculator in radian mode, and then we obtain

$$\sin 3 \approx 0.14112$$

If we want to know the sine of the angle 3° we would write $\sin 3^\circ$ and, with our calculator in degree mode, we find that

$$\sin 3^\circ \approx 0.05234$$

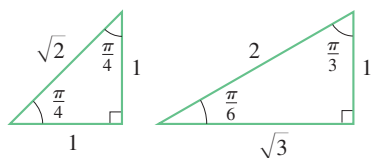


FIGURE 8

The exact trigonometric ratios for certain angles can be read from the triangles in Figure 8. For instance,

$$\begin{array}{lll} \sin \frac{\pi}{4} = \frac{1}{\sqrt{2}} & \sin \frac{\pi}{6} = \frac{1}{2} & \sin \frac{\pi}{3} = \frac{\sqrt{3}}{2} \\ \cos \frac{\pi}{4} = \frac{1}{\sqrt{2}} & \cos \frac{\pi}{6} = \frac{\sqrt{3}}{2} & \cos \frac{\pi}{3} = \frac{1}{2} \\ \tan \frac{\pi}{4} = 1 & \tan \frac{\pi}{6} = \frac{1}{\sqrt{3}} & \tan \frac{\pi}{3} = \sqrt{3} \end{array}$$

The signs of the trigonometric functions for angles in each of the four quadrants can be remembered by means of the rule “All Students Take Calculus” shown in Figure 9.

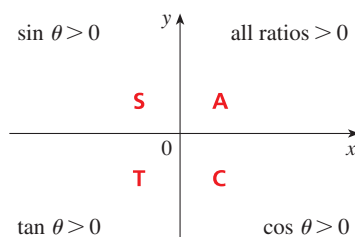


FIGURE 9

EXAMPLE 3 | Find the exact trigonometric ratios for $\theta = 2\pi/3$.

SOLUTION From Figure 10 we see that a point on the terminal line for $\theta = 2\pi/3$ is $P(-1, \sqrt{3})$. Therefore, taking

$$x = -1 \quad y = \sqrt{3} \quad r = 2$$

in the definitions of the trigonometric ratios, we have

$$\begin{array}{lll} \sin \frac{2\pi}{3} = \frac{\sqrt{3}}{2} & \cos \frac{2\pi}{3} = -\frac{1}{2} & \tan \frac{2\pi}{3} = -\sqrt{3} \\ \csc \frac{2\pi}{3} = \frac{2}{\sqrt{3}} & \sec \frac{2\pi}{3} = -2 & \cot \frac{2\pi}{3} = -\frac{1}{\sqrt{3}} \end{array}$$

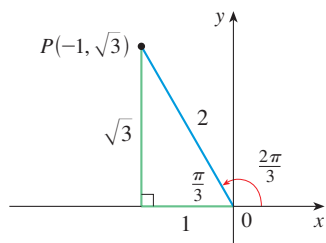


FIGURE 10

The following table gives some values of $\sin \theta$ and $\cos \theta$ found by the method of Example 3.

θ	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$	$\frac{2\pi}{3}$	$\frac{3\pi}{4}$	$\frac{5\pi}{6}$	π	$\frac{3\pi}{2}$	2π
$\sin \theta$	0	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0	-1	0
$\cos \theta$	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0	$-\frac{1}{2}$	$-\frac{1}{\sqrt{2}}$	$-\frac{\sqrt{3}}{2}$	-1	0	1

EXAMPLE 4 | If $\cos \theta = \frac{2}{5}$ and $0 < \theta < \pi/2$, find the other five trigonometric functions of θ .

SOLUTION Since $\cos \theta = \frac{2}{5}$, we can label the hypotenuse as having length 5 and the adjacent side as having length 2 in Figure 11. If the opposite side has length x , then the

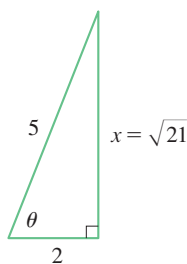


FIGURE 11

Pythagorean Theorem gives $x^2 + 4 = 25$ and so $x^2 = 21$, $x = \sqrt{21}$. We can now use the diagram to write the other five trigonometric functions:

$$\begin{aligned}\sin \theta &= \frac{\sqrt{21}}{5} & \tan \theta &= \frac{\sqrt{21}}{2} \\ \csc \theta &= \frac{5}{\sqrt{21}} & \sec \theta &= \frac{5}{2} & \cot \theta &= \frac{2}{\sqrt{21}}\end{aligned}$$

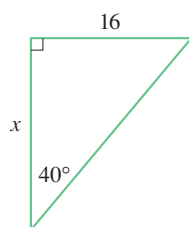


FIGURE 12

EXAMPLE 5 | Use a calculator to approximate the value of x in Figure 12.

SOLUTION From the diagram we see that

$$\tan 40^\circ = \frac{16}{x}$$

Therefore

$$x = \frac{16}{\tan 40^\circ} \approx 19.07$$

Trigonometric Identities

A trigonometric identity is a relationship among the trigonometric functions. The most elementary are the following, which are immediate consequences of the definitions of the trigonometric functions.

$$\begin{aligned}(6) \quad \csc \theta &= \frac{1}{\sin \theta} & \sec \theta &= \frac{1}{\cos \theta} & \cot \theta &= \frac{1}{\tan \theta} \\ \tan \theta &= \frac{\sin \theta}{\cos \theta} & \cot \theta &= \frac{\cos \theta}{\sin \theta}\end{aligned}$$

For the next identity we refer back to Figure 7. The distance formula (or, equivalently, the Pythagorean Theorem) tells us that $x^2 + y^2 = r^2$. Therefore

$$\sin^2 \theta + \cos^2 \theta = \frac{y^2}{r^2} + \frac{x^2}{r^2} = \frac{x^2 + y^2}{r^2} = \frac{r^2}{r^2} = 1$$

We have therefore proved one of the most useful of all trigonometric identities:

$$(7) \quad \sin^2 \theta + \cos^2 \theta = 1$$

If we now divide both sides of Equation 7 by $\cos^2 \theta$ and use Equations 6, we get

$$(8) \quad \tan^2 \theta + 1 = \sec^2 \theta$$

Similarly, if we divide both sides of Equation 7 by $\sin^2 \theta$, we get

$$(9) \quad 1 + \cot^2 \theta = \csc^2 \theta$$

The identities

(10a)

(10b)

$$\sin(-\theta) = -\sin \theta$$

$$\cos(-\theta) = \cos \theta$$

Odd functions and even functions are discussed in Section 1.1.

show that sine is an odd function and cosine is an even function. They are easily proved by drawing a diagram showing θ and $-\theta$ in standard position (see Exercise 19).

Since the angles θ and $\theta + 2\pi$ have the same terminal side, we have

(11)

$$\sin(\theta + 2\pi) = \sin \theta \quad \cos(\theta + 2\pi) = \cos \theta$$

These identities show that the sine and cosine functions are periodic with period 2π .

The remaining trigonometric identities are all consequences of two basic identities called the **addition formulas**:

(12a)

(12b)

$$\sin(x + y) = \sin x \cos y + \cos x \sin y$$

$$\cos(x + y) = \cos x \cos y - \sin x \sin y$$

The proofs of these addition formulas are outlined in Exercises 45, 46, and 47.

By substituting $-y$ for y in Equations 12a and 12b and using Equations 10a and 10b, we obtain the following **subtraction formulas**:

(13a)

(13b)

$$\sin(x - y) = \sin x \cos y - \cos x \sin y$$

$$\cos(x - y) = \cos x \cos y + \sin x \sin y$$

Then, by dividing the formulas in Equations 12 or Equations 13, we obtain the corresponding formulas for $\tan(x \pm y)$:

(14a)

(14b)

$$\tan(x + y) = \frac{\tan x + \tan y}{1 - \tan x \tan y}$$

$$\tan(x - y) = \frac{\tan x - \tan y}{1 + \tan x \tan y}$$

If we put $y = x$ in the addition formulas (12), we get the **double-angle formulas**:

(15a)

(15b)

$$\sin 2x = 2 \sin x \cos x$$

$$\cos 2x = \cos^2 x - \sin^2 x$$

Then, by using the identity $\sin^2 x + \cos^2 x = 1$, we obtain the following alternate forms of the double-angle formulas for $\cos 2x$:

(16a)

$$\cos 2x = 2 \cos^2 x - 1$$

(16b)

$$\cos 2x = 1 - 2 \sin^2 x$$

If we now solve these equations for $\cos^2 x$ and $\sin^2 x$, we get the following **half-angle formulas**, which are useful in integral calculus:

(17a)

$$\cos^2 x = \frac{1 + \cos 2x}{2}$$

(17b)

$$\sin^2 x = \frac{1 - \cos 2x}{2}$$

Finally, we state the **product formulas**, which can be deduced from Equations 12 and 13:

(18a)

$$\sin x \cos y = \frac{1}{2}[\sin(x + y) + \sin(x - y)]$$

(18b)

$$\cos x \cos y = \frac{1}{2}[\cos(x + y) + \cos(x - y)]$$

(18c)

$$\sin x \sin y = \frac{1}{2}[\cos(x - y) - \cos(x + y)]$$

There are many other trigonometric identities, but those we have stated are the ones used most often in calculus. If you forget any of them, remember that they can all be deduced from Equations 12a and 12b.

EXAMPLE 6 | Find all values of x in the interval $[0, 2\pi]$ such that $\sin x = \sin 2x$.

SOLUTION Using the double-angle formula (15a), we rewrite the given equation as

$$\sin x = 2 \sin x \cos x \quad \text{or} \quad \sin x(1 - 2 \cos x) = 0$$

Therefore there are two possibilities:

$$\sin x = 0 \quad \text{or} \quad 1 - 2 \cos x = 0$$

$$x = 0, \pi, 2\pi \quad \cos x = \frac{1}{2}$$

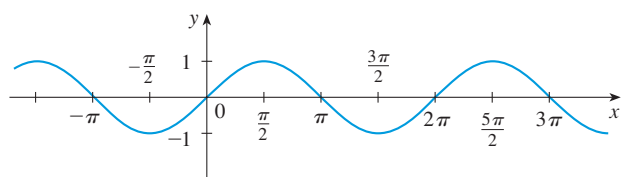
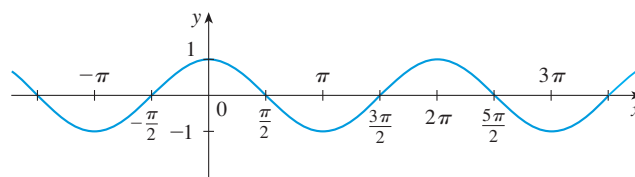
$$x = \frac{\pi}{3}, \frac{5\pi}{3}$$

The given equation has five solutions: $0, \pi/3, \pi, 5\pi/3$, and 2π . ■

■ Graphs of the Trigonometric Functions

The graph of the function $f(x) = \sin x$, shown in Figure 13(a) on page 706, is obtained by plotting points for $0 \leq x \leq 2\pi$ and then using the periodic nature of the function (from Equation 11) to complete the graph. Notice that the zeros of the sine function occur at the integer multiples of π , that is,

$$\sin x = 0 \quad \text{whenever } x = n\pi, \quad n \text{ an integer}$$

(a) $f(x) = \sin x$ (b) $g(x) = \cos x$ **FIGURE 13**

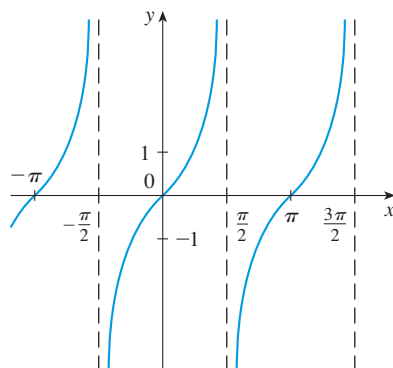
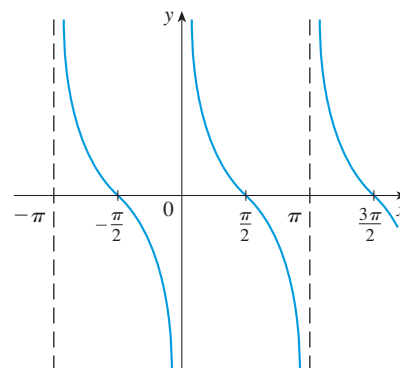
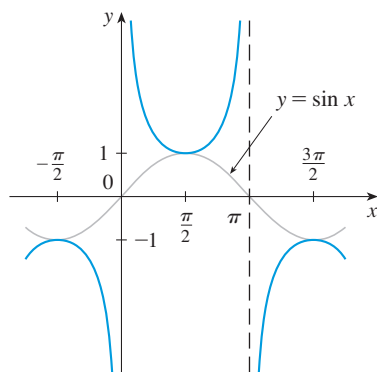
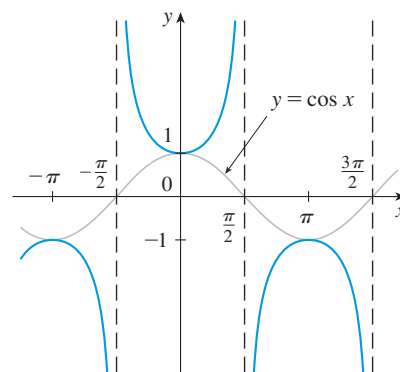
Because of the identity

$$\cos x = \sin\left(x + \frac{\pi}{2}\right)$$

(which can be verified using Equation 12a), the graph of cosine is obtained by shifting the graph of sine by an amount $\pi/2$ to the left [see Figure 13(b)]. Note that for both the sine and cosine functions the domain is $(-\infty, \infty)$ and the range is the closed interval $[-1, 1]$. Thus, for all values of x , we have

$$-1 \leq \sin x \leq 1 \quad -1 \leq \cos x \leq 1$$

The graphs of the remaining four trigonometric functions are shown in Figure 14 and their domains are indicated there. Notice that tangent and cotangent have range $(-\infty, \infty)$, whereas cosecant and secant have range $(-\infty, -1] \cup [1, \infty)$. All four functions are periodic: tangent and cotangent have period π , whereas cosecant and secant have period 2π .

(a) $y = \tan x$ (b) $y = \cot x$ (c) $y = \csc x$ (d) $y = \sec x$ **FIGURE 14**

EXERCISES C

1–2 Convert from degrees to radians.

1. (a) 210° (b) 9°
 2. (a) -315° (b) 36°

3–4 Convert from radians to degrees.

3. (a) 4π (b) $-\frac{3\pi}{8}$
 4. (a) $-\frac{7\pi}{2}$ (b) $\frac{8\pi}{3}$

5. Find the length of a circular arc subtended by an angle of $\pi/12$ rad if the radius of the circle is 36 cm.
 6. If a circle has radius 10 cm, find the length of the arc subtended by a central angle of 72° .
 7. A circle has radius 1.5 m. What angle is subtended at the center of the circle by an arc 1 m long?
 8. Find the radius of a circular sector with angle $3\pi/4$ and arc length 6 cm.

9–10 Draw, in standard position, the angle whose measure is given.

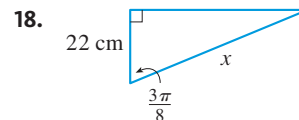
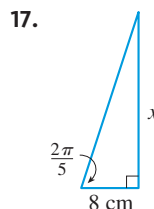
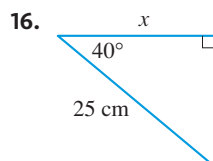
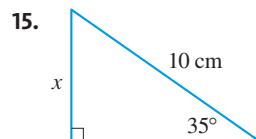
9. (a) 315° (b) $-\frac{3\pi}{4}$ rad
 10. (a) $\frac{7\pi}{3}$ rad (b) -3 rad

11–12 Find the exact trigonometric ratios for the angle whose radian measure is given.

11. $\frac{3\pi}{4}$ 12. $\frac{4\pi}{3}$

13–14 Find the remaining trigonometric ratios.

13. $\sin \theta = \frac{3}{5}$, $0 < \theta < \frac{\pi}{2}$
 14. $\tan \alpha = 2$, $0 < \alpha < \frac{\pi}{2}$

15–18 Find, correct to five decimal places, the length of the side labeled x .**19–21** Prove each equation.

19. (a) Equation 10a (b) Equation 10b
 20. (a) Equation 14a (b) Equation 14b
 21. (a) Equation 18a (b) Equation 18b
 (c) Equation 18c

22–28 Prove the identity.

22. $\cos\left(\frac{\pi}{2} - x\right) = \sin x$
 23. $\sin\left(\frac{\pi}{2} + x\right) = \cos x$ 24. $\sin(\pi - x) = \sin x$
 25. $\sin \theta \cot \theta = \cos \theta$ 26. $(\sin x + \cos x)^2 = 1 + \sin 2x$
 27. $\tan 2\theta = \frac{2 \tan \theta}{1 - \tan^2 \theta}$ 28. $\cos 3\theta = 4 \cos^3 \theta - 3 \cos \theta$

29–30 If $\sin x = \frac{1}{3}$ and $\sec y = \frac{5}{4}$, where x and y lie between 0 and $\pi/2$, evaluate the expression.

29. $\sin(x + y)$ 30. $\cos 2y$

31–34 Find all values of x in the interval $[0, 2\pi]$ that satisfy the equation.

31. $2 \cos x - 1 = 0$ 32. $2 \sin^2 x = 1$
 33. $\sin 2x = \cos x$ 34. $|\tan x| = 1$

35–38 Find all values of x in the interval $[0, 2\pi]$ that satisfy the inequality.

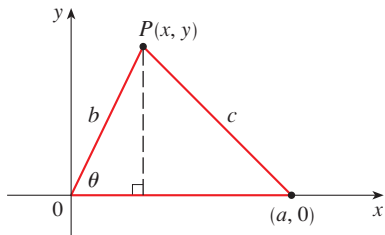
35. $\sin x \leq \frac{1}{2}$ 36. $2 \cos x + 1 > 0$
 37. $-1 < \tan x < 1$ 38. $\sin x > \cos x$

39–42 Graph the function by starting with the graphs in Figures 13 and 14 and applying the transformations of Section 1.3 where appropriate.

39. $y = \cos\left(x - \frac{\pi}{3}\right)$ 40. $y = \tan 2x$
 41. $y = \frac{1}{3} \tan\left(x - \frac{\pi}{2}\right)$ 42. $y = |\sin x|$

43. Prove the **Law of Cosines**: If a triangle has sides with lengths a , b , and c , and θ is the angle between the sides with lengths a and b , then

$$c^2 = a^2 + b^2 - 2ab \cos \theta$$

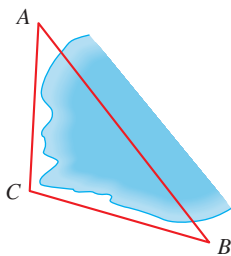


[Hint: Introduce a coordinate system so that θ is in standard position, as in the figure. Express x and y in terms of θ and then use the distance formula to compute c .]

44. In order to find the distance $|AB|$ across a small inlet, a point C is located as in the figure and the following measurements were recorded:

$$\angle C = 103^\circ \quad |AC| = 820 \text{ m} \quad |BC| = 910 \text{ m}$$

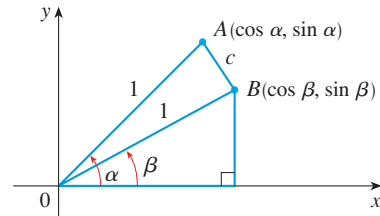
Use the Law of Cosines from Exercise 43 to find the required distance.



45. Use the figure to prove the subtraction formula

$$\cos(\alpha - \beta) = \cos \alpha \cos \beta + \sin \alpha \sin \beta$$

[Hint: Compute c^2 in two ways (using the Law of Cosines from Exercise 43 and also using the distance formula) and compare the two expressions.]



46. Use the formula in Exercise 45 to prove the addition formula for cosine (12b).

47. Use the addition formula for cosine and the identities

$$\cos\left(\frac{\pi}{2} - \theta\right) = \sin \theta \quad \sin\left(\frac{\pi}{2} - \theta\right) = \cos \theta$$

to prove the subtraction formula for the sine function.

48. (a) Show that the area of a triangle with sides of lengths a and b and with included angle θ is

$$A = \frac{1}{2}ab \sin \theta$$

- (b) Find the area of triangle ABC , correct to five decimal places, if

$$|AB| = 10 \text{ cm} \quad |BC| = 3 \text{ cm} \quad \angle ABC = 107^\circ$$

D Precise Definitions of Limits

The definitions of limits that have been given in this book are appropriate for intuitive understanding of the basic concepts of calculus. For the purposes of deeper understanding and rigorous proofs, however, the precise definitions of this appendix are necessary. In particular, the definition of a limit given here is used in Appendix E to prove some of the properties of limits.

Sequences

In Section 2.1 we used the notation $\lim_{n \rightarrow \infty} a_n = L$ to mean that the terms of the sequence $\{a_n\}$ approach the number L as n becomes large. Here we want to express, in quantitative form, that a_n can be made arbitrarily close to L by making n sufficiently large. This means that a_n can be made to lie within a prescribed distance from L (traditionally denoted by ϵ , the Greek letter epsilon) by requiring that n be made larger than a specified number N . The resulting precise definition of a limit is as follows.

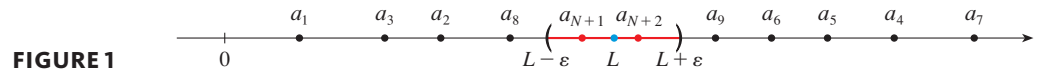
(1) Definition A sequence $\{a_n\}$ has the **limit** L and we write

$$\lim_{n \rightarrow \infty} a_n = L \quad \text{or} \quad a_n \rightarrow L \quad \text{as} \quad n \rightarrow \infty$$

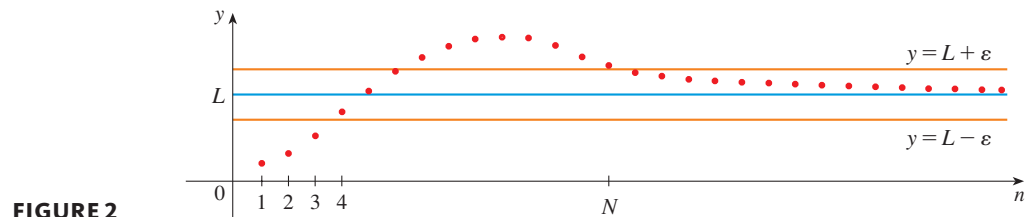
if for every $\varepsilon > 0$ there is a corresponding number N such that

$$\text{if } n > N \quad \text{then} \quad |a_n - L| < \varepsilon$$

Definition 1 is illustrated by Figure 1, in which the terms $a_1, a_2, a_3, \dots$ are plotted on a number line. No matter how small an interval $(L - \varepsilon, L + \varepsilon)$ is chosen, there exists an N such that all terms of the sequence from a_{N+1} onward must lie in that interval.



Another illustration of Definition 1 is given in Figure 2. The points on the graph of $\{a_n\}$ must lie between the horizontal lines $y = L + \varepsilon$ and $y = L - \varepsilon$ if $n > N$. This picture must be valid no matter how small ε is chosen, but usually a smaller ε requires a larger N .



EXAMPLE 1 | Prove that $\lim_{n \rightarrow \infty} \frac{1}{n} = 0$.

SOLUTION Given $\varepsilon > 0$, we want to find a number N such that

$$\text{if } n > N \quad \text{then} \quad \left| \frac{1}{n} - 0 \right| < \varepsilon$$

But $1/n < \varepsilon \iff n > 1/\varepsilon$. Let's choose $N = 1/\varepsilon$. So

$$\text{if } n > N = \frac{1}{\varepsilon} \quad \text{then} \quad \left| \frac{1}{n} - 0 \right| = \frac{1}{n} < \varepsilon$$

Therefore, by Definition 1,

$$\lim_{n \rightarrow \infty} \frac{1}{n} = 0$$

Functions

The precise definition of the limit of a function at infinity, $\lim_{x \rightarrow \infty} f(x) = L$, is very similar to the definition of the limit of a sequence, $\lim_{n \rightarrow \infty} a_n = L$, given in Definition 1. So let's turn our attention to the definition of the limit of $f(x)$ as x approaches a finite number.

For a precise definition of $\lim_{x \rightarrow a} f(x) = L$ we want to express that $f(x)$ can be made arbitrarily close to L by making x sufficiently close to a (but $x \neq a$). This means that $f(x)$ can be made to lie within any preassigned distance ε from L by requiring that x be within a specified distance δ (the Greek letter delta) from a . That is, $|f(x) - L| < \varepsilon$ when

$|x - a| < \delta$ and $x \neq a$. Notice that we can stipulate that $x \neq a$ by writing $0 < |x - a|$. The resulting precise definition of a limit is as follows.

(2) Definition Let f be a function defined on some open interval that contains the number a , except possibly at a itself. Then we say that the **limit of $f(x)$ as x approaches a is L** , and we write

$$\lim_{x \rightarrow a} f(x) = L$$

if for every number $\varepsilon > 0$ there is a number $\delta > 0$ such that

$$\text{if } 0 < |x - a| < \delta \quad \text{then} \quad |f(x) - L| < \varepsilon$$

Definition 2 is illustrated in Figures 3–5. If a number $\varepsilon > 0$ is given, then we draw the horizontal lines $y = L + \varepsilon$ and $y = L - \varepsilon$ and the graph of f . (See Figure 3.) If $\lim_{x \rightarrow a} f(x) = L$, then we can find a number $\delta > 0$ such that if we restrict x to lie in the interval $(a - \delta, a + \delta)$ and take $x \neq a$, then the curve $y = f(x)$ lies between the lines $y = L - \varepsilon$ and $y = L + \varepsilon$. (See Figure 4.) You can see that if such a δ has been found, then any smaller δ will also work.

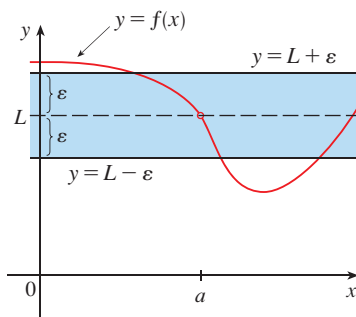


FIGURE 3

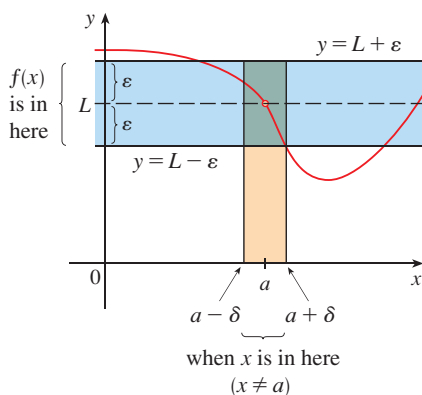


FIGURE 4

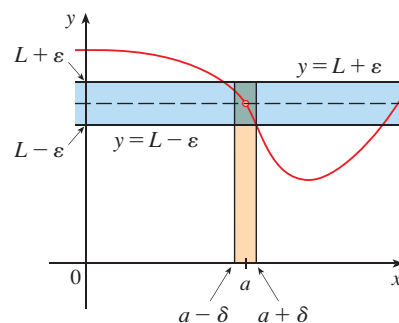


FIGURE 5

TEC In Module D you can explore the precise definition of a limit both graphically and numerically.

It's important to realize that the process illustrated in Figures 3 and 4 must work for every positive number ε no matter how small it is chosen. Figure 5 shows that if a smaller ε is chosen, then a smaller δ may be required.

EXAMPLE 2 | Use the ε, δ definition to prove that $\lim_{x \rightarrow 0} x^2 = 0$.

SOLUTION Let ε be a given positive number. According to Definition 2 with $a = 0$ and $L = 0$, we need to find a number δ such that

$$\text{if } 0 < |x - 0| < \delta \quad \text{then} \quad |x^2 - 0| < \varepsilon$$

$$\text{that is,} \quad \text{if } 0 < |x| < \delta \quad \text{then} \quad x^2 < \varepsilon$$

But, since the square root function is an increasing function, we know that

$$x^2 < \varepsilon \iff \sqrt{x^2} < \sqrt{\varepsilon} \iff |x| < \sqrt{\varepsilon}$$

So if we choose $\delta = \sqrt{\varepsilon}$, then $x^2 < \varepsilon \iff |x| < \delta$. (See Figure 6.) This shows that $\lim_{x \rightarrow 0} x^2 = 0$.

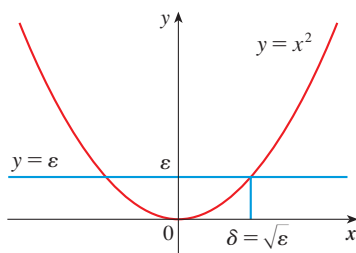


FIGURE 6

In proving limit statements it may be helpful to think of the definition of a limit as a challenge. First it challenges you with a number ε . Then you must be able to produce a suitable δ . You have to be able to do this for *every* $\varepsilon > 0$, not just a particular ε .

EXAMPLE 3 | Prove that $\lim_{x \rightarrow 3} (4x - 5) = 7$.

SOLUTION Let ε be a given positive number. According to Definition 2 with $a = 3$ and $L = 7$, we need to find a number δ such that

$$\text{if } 0 < |x - 3| < \delta \quad \text{then} \quad |(4x - 5) - 7| < \varepsilon$$

But $|(4x - 5) - 7| = |4x - 12| = |4(x - 3)| = 4|x - 3|$. Therefore we want δ such that

$$\text{if } 0 < |x - 3| < \delta \quad \text{then} \quad 4|x - 3| < \varepsilon$$

Note that $4|x - 3| < \varepsilon \iff |x - 3| < \varepsilon/4$. So let's choose $\delta = \varepsilon/4$. We can then write the following:

$$\text{If } 0 < |x - 3| < \delta \quad \text{then} \quad 4|x - 3| < \varepsilon \quad \text{so} \quad |(4x - 5) - 7| < \varepsilon$$

Therefore, by the definition of a limit,

$$\lim_{x \rightarrow 3} (4x - 5) = 7$$

Figure 7 shows the geometry behind Example 3.

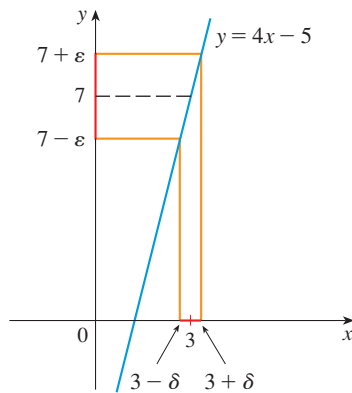


FIGURE 7

This example is illustrated by Figure 7.

It's not always easy to prove that limit statements are true using the ε, δ definition. For a more complicated function such as $f(x) = (6x^2 - 8x + 9)/(2x^2 - 1)$, a proof would require a great deal of ingenuity. Fortunately, this is not necessary because the Limit Laws stated in Section 2.4 can be proved using Definition 2, and then the limits of complicated functions can be found rigorously from the Limit Laws without resorting to the definition directly.

Definite Integrals

In Section 5.2 we defined the definite integral of a function f on an interval $[a, b]$ as

$$\int_a^b f(x) dx = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i^*) \Delta x$$

where, at the n th stage, we have divided $[a, b]$ into n subintervals of equal width, $\Delta x = (b - a)/n$, and x_i^* is any sample point in the i th subinterval. The precise meaning of this limit that defines the integral is as follows:

For every number $\varepsilon > 0$ there is an integer N such that

$$\left| \int_a^b f(x) dx - \sum_{i=1}^n f(x_i^*) \Delta x \right| < \varepsilon$$

for every integer $n > N$ and for every choice of x_i^* in the i th subinterval.

This means that a definite integral can be approximated to within any desired degree of accuracy by a Riemann sum.

■ Functions of Two Variables

Here is a precise version of Definition 9.1.1:

(3) Definition Let f be a function of two variables whose domain D includes points arbitrarily close to (a, b) . Then we say that the **limit of $f(x, y)$ as (x, y) approaches (a, b) is L** and we write

$$\lim_{(x, y) \rightarrow (a, b)} f(x, y) = L$$

if for every number $\varepsilon > 0$ there is a corresponding number $\delta > 0$ such that

$$\text{if } (x, y) \in D \quad \text{and} \quad 0 < \sqrt{(x - a)^2 + (y - b)^2} < \delta$$

$$\text{then} \quad |f(x, y) - L| < \varepsilon$$

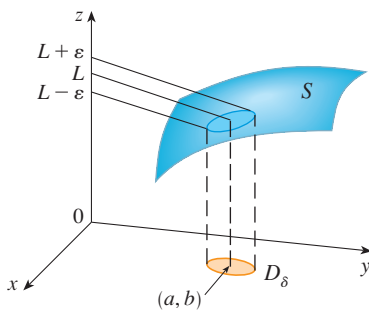


FIGURE 8

Notice that $|f(x, y) - L|$ is the distance between the numbers $f(x, y)$ and L , and $\sqrt{(x - a)^2 + (y - b)^2}$ is the distance between the point (x, y) and the point (a, b) . Thus Definition 3 says that the distance between $f(x, y)$ and L can be made arbitrarily small by making the distance from (x, y) to (a, b) sufficiently small (but not 0). An illustration of Definition 3 is given in Figure 8 where the surface S is the graph of f . If $\varepsilon > 0$ is given, we can find $\delta > 0$ such that if (x, y) is restricted to lie in the disk D_δ with center (a, b) and radius δ , and if $(x, y) \neq (a, b)$, then the corresponding part of S lies between the horizontal planes $z = L - \varepsilon$ and $z = L + \varepsilon$.

EXAMPLE 4 | Prove that $\lim_{(x, y) \rightarrow (0, 0)} \frac{3x^2y}{x^2 + y^2} = 0$.

SOLUTION Let $\varepsilon > 0$. We want to find $\delta > 0$ such that

$$\text{if } 0 < \sqrt{x^2 + y^2} < \delta \quad \text{then} \quad \left| \frac{3x^2y}{x^2 + y^2} - 0 \right| < \varepsilon$$

$$\text{that is,} \quad \text{if } 0 < \sqrt{x^2 + y^2} < \delta \quad \text{then} \quad \frac{3x^2|y|}{x^2 + y^2} < \varepsilon$$

But $x^2 \leq x^2 + y^2$ since $y^2 \geq 0$, so $x^2/(x^2 + y^2) \leq 1$ and therefore

$$\frac{3x^2|y|}{x^2 + y^2} \leq 3|y| = 3\sqrt{y^2} \leq 3\sqrt{x^2 + y^2}$$

Thus if we choose $\delta = \varepsilon/3$ and let $0 < \sqrt{x^2 + y^2} < \delta$, then

$$\left| \frac{3x^2y}{x^2 + y^2} - 0 \right| \leq 3\sqrt{x^2 + y^2} < 3\delta = 3\left(\frac{\varepsilon}{3}\right) = \varepsilon$$

Hence, by Definition 3,

$$\lim_{(x, y) \rightarrow (0, 0)} \frac{3x^2y}{x^2 + y^2} = 0$$

EXERCISES D

1. (a) Determine how large we have to take n so that

$$\frac{1}{n^2} < 0.0001$$

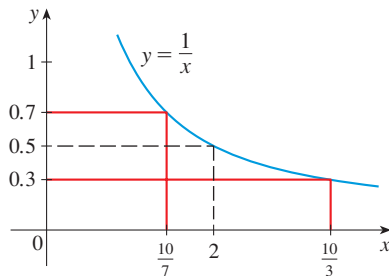
- (b) Use Definition 1 to prove that

$$\lim_{n \rightarrow \infty} \frac{1}{n^2} = 0$$

2. Use Definition 1 to prove that $\lim_{n \rightarrow \infty} \frac{1}{\sqrt{n}} = 0$.

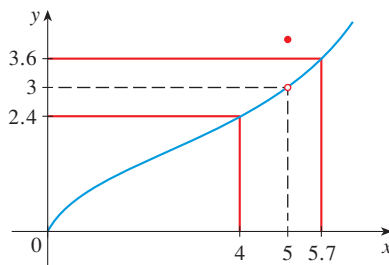
3. Use the given graph of $f(x) = 1/x$ to find a number δ such that

$$\text{if } |x - 2| < \delta \quad \text{then} \quad \left| \frac{1}{x} - 0.5 \right| < 0.2$$



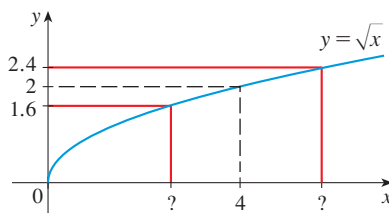
4. Use the given graph of f to find a number δ such that

$$\text{if } 0 < |x - 5| < \delta \quad \text{then} \quad |f(x) - 3| < 0.6$$



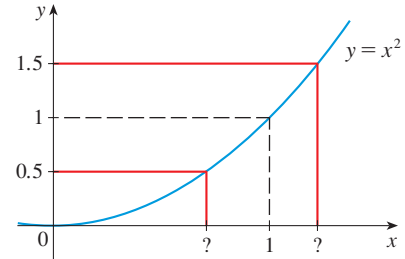
5. Use the given graph of $f(x) = \sqrt{x}$ to find a number δ such that

$$\text{if } |x - 4| < \delta \quad \text{then} \quad |\sqrt{x} - 2| < 0.4$$



6. Use the given graph of $f(x) = x^2$ to find a number δ such that

$$\text{if } |x - 1| < \delta \quad \text{then} \quad |x^2 - 1| < \frac{1}{2}$$



7. Use a graph to find a number δ such that

$$\text{if } \left| x - \frac{\pi}{4} \right| < \delta \quad \text{then} \quad |\tan x - 1| < 0.2$$

8. Use a graph to find a number δ such that

$$\text{if } |x - 1| < \delta \quad \text{then} \quad \left| \frac{2x}{x^2 + 4} - 0.4 \right| < 0.1$$

9. For the limit

$$\lim_{x \rightarrow 1} (4 + x - 3x^3) = 2$$

illustrate Definition 2 by finding values of δ that correspond to $\varepsilon = 1$ and $\varepsilon = 0.1$.

10. For the limit

$$\lim_{x \rightarrow 0} \frac{e^x - 1}{x} = 1$$

illustrate Definition 2 by finding values of δ that correspond to $\varepsilon = 0.5$ and $\varepsilon = 0.1$.

11. Use Definition 2 to prove that $\lim_{x \rightarrow 0} x^3 = 0$.

12. (a) How would you formulate an ε, δ definition of the one-sided limit $\lim_{x \rightarrow a^+} f(x) = L$?

- (b) Use your definition in part (a) to prove that $\lim_{x \rightarrow 0^+} \sqrt{x} = 0$.

13. A machinist is required to manufacture a circular metal disk with area 1000 cm^2 .

- (a) What radius produces such a disk?

- (b) If the machinist is allowed an error tolerance of $\pm 5 \text{ cm}^2$ in the area of the disk, how close to the ideal radius in part (a) must the machinist control the radius?

- (c) In terms of the ε, δ definition of $\lim_{x \rightarrow a} f(x) = L$, what is x ? What is $f(x)$? What is a ? What is L ? What value of ε is given? What is the corresponding value of δ ?

14. A crystal growth furnace is used in research to determine how best to manufacture crystals used in electronic compo-

nents for the space shuttle. For proper growth of the crystal, the temperature must be controlled accurately by adjusting the input power. Suppose the relationship is given by

$$T(w) = 0.1w^2 + 2.155w + 20$$

where T is the temperature in degrees Celsius and w is the power input in watts.

- (a) How much power is needed to maintain the temperature at 200°C ?
 - (b) If the temperature is allowed to vary from 200°C by up to $\pm 1^\circ\text{C}$, what range of wattage is allowed for the input power?
 - (c) In terms of the ε, δ definition of $\lim_{x \rightarrow a} f(x) = L$, what is x ? What is $f(x)$? What is a ? What is L ? What value of ε is given? What is the corresponding value of δ ?
15. (a) Find a number δ such that if $|x - 2| < \delta$, then $|4x - 8| < \varepsilon$, where $\varepsilon = 0.1$.
(b) Repeat part (a) with $\varepsilon = 0.01$.
16. Given that $\lim_{x \rightarrow 2} (5x - 7) = 3$, illustrate Definition 2 by finding values of δ that correspond to $\varepsilon = 0.1$, $\varepsilon = 0.05$, and $\varepsilon = 0.01$.

17–18 Prove the statement using the ε, δ definition of a limit and illustrate with a diagram like Figure 7.

17. $\lim_{x \rightarrow -3} (1 - 4x) = 13$

18. $\lim_{x \rightarrow -2} (\frac{1}{2}x + 3) = 2$

19. Use Definition 1 to prove that

(a) $\lim_{n \rightarrow \infty} \frac{1}{2^n} = 0$

(b) $\lim_{n \rightarrow \infty} r^n = 0$ if $|r| < 1$

20. (a) For what values of x is it true that

$$\frac{1}{x^2} > 1,000,000$$

(b) The precise definition of $\lim_{x \rightarrow a} f(x) = \infty$ states that for every positive number M (no matter how large) there is a corresponding positive number δ such that if $0 < |x - a| < \delta$, then $f(x) > M$. Use this definition to prove that $\lim_{x \rightarrow 0} (1/x^2) = \infty$.

21. Use Definition 3 to prove that $\lim_{(x,y) \rightarrow (0,0)} \frac{xy}{\sqrt{x^2 + y^2}} = 0$.

E | A Few Proofs

In this appendix we present proofs of some theorems that were stated in the main body of the text. We start by proving the Triangle Inequality, which is an important property of absolute value.

The Triangle Inequality If a and b are any real numbers, then

$$|a + b| \leq |a| + |b|$$

Observe that if the numbers a and b are both positive or both negative, then the two sides in the Triangle Inequality are actually equal. But if a and b have opposite signs, the left side involves a subtraction and the right side does not. This makes the Triangle Inequality seem reasonable, but we can prove it as follows.

Notice that

$$-|a| \leq a \leq |a|$$

is always true because a equals either $|a|$ or $-|a|$. The corresponding statement for b is

$$-|b| \leq b \leq |b|$$

Adding these inequalities, we get

$$-(|a| + |b|) \leq a + b \leq |a| + |b|$$

If we now apply Properties 4 and 5 of absolute value from appendix A (with x replaced by $a + b$ and a by $|a| + |b|$), we obtain

$$|a + b| \leq |a| + |b|$$

which is what we wanted to show.

When combined, Properties 4 and 5 of absolute value (see Appendix A) say that

$$|x| \leq a \iff -a \leq x \leq a$$

The Sum Law was first stated in Section 2.4.

Next we use the Triangle Inequality to prove the Sum Law for limits.

Sum Law If $\lim_{x \rightarrow a} f(x) = L$ and $\lim_{x \rightarrow a} g(x) = M$ both exist, then

$$\lim_{x \rightarrow a} [f(x) + g(x)] = L + M$$

PROOF Let $\varepsilon > 0$ be given. According to Definition 2 in Appendix D, we must find $\delta > 0$ such that

$$\text{if } 0 < |x - a| < \delta \quad \text{then} \quad |f(x) + g(x) - (L + M)| < \varepsilon$$

Using the Triangle Inequality we can write

$$\begin{aligned} (1) \quad |f(x) + g(x) - (L + M)| &= |(f(x) - L) + (g(x) - M)| \\ &\leq |f(x) - L| + |g(x) - M| \end{aligned}$$

We will make $|f(x) + g(x) - (L + M)|$ less than ε by making each of the terms $|f(x) - L|$ and $|g(x) - M|$ less than $\varepsilon/2$.

Since $\varepsilon/2 > 0$ and $\lim_{x \rightarrow a} f(x) = L$, there exists a number $\delta_1 > 0$ such that

$$\text{if } 0 < |x - a| < \delta_1 \quad \text{then} \quad |f(x) - L| < \frac{\varepsilon}{2}$$

Similarly, since $\lim_{x \rightarrow a} g(x) = M$, there exists a number $\delta_2 > 0$ such that

$$\text{if } 0 < |x - a| < \delta_2 \quad \text{then} \quad |g(x) - M| < \frac{\varepsilon}{2}$$

Let $\delta = \min\{\delta_1, \delta_2\}$, the smaller of the numbers δ_1 and δ_2 . Notice that

$$\text{if } 0 < |x - a| < \delta \quad \text{then} \quad 0 < |x - a| < \delta_1 \quad \text{and} \quad 0 < |x - a| < \delta_2$$

$$\text{and so} \quad |f(x) - L| < \frac{\varepsilon}{2} \quad \text{and} \quad |g(x) - M| < \frac{\varepsilon}{2}$$

Therefore, by (1),

$$\begin{aligned} |f(x) + g(x) - (L + M)| &\leq |f(x) - L| + |g(x) - M| \\ &< \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon \end{aligned}$$

To summarize,

$$\text{if } 0 < |x - a| < \delta \quad \text{then} \quad |f(x) + g(x) - (L + M)| < \varepsilon$$

Thus, by the definition of a limit,

$$\lim_{x \rightarrow a} [f(x) + g(x)] = L + M$$

Fermat's Theorem was discussed in Section 4.1.

Fermat's Theorem If f has a local maximum or minimum at c , and if $f'(c)$ exists, then $f'(c) = 0$.

PROOF Suppose, for the sake of definiteness, that f has a local maximum at c . Then, $f(c) \geq f(x)$ if x is sufficiently close to c . This implies that if h is sufficiently close to 0,

with h being positive or negative, then

$$f(c) \geq f(c + h)$$

and therefore

$$(2) \quad f(c + h) - f(c) \leq 0$$

We can divide both sides of an inequality by a positive number. Thus, if $h > 0$ and h is sufficiently small, we have

$$\frac{f(c + h) - f(c)}{h} \leq 0$$

Taking the right-hand limit of both sides of this inequality (using Theorem 2.4.2), we get

$$\lim_{h \rightarrow 0^+} \frac{f(c + h) - f(c)}{h} \leq \lim_{h \rightarrow 0^+} 0 = 0$$

But since $f'(c)$ exists, we have

$$f'(c) = \lim_{h \rightarrow 0} \frac{f(c + h) - f(c)}{h} = \lim_{h \rightarrow 0^+} \frac{f(c + h) - f(c)}{h}$$

and so we have shown that $f'(c) \leq 0$.

If $h < 0$, then the direction of the inequality (2) is reversed when we divide by h :

$$\frac{f(c + h) - f(c)}{h} \geq 0 \quad h < 0$$

So, taking the left-hand limit, we have

$$f'(c) = \lim_{h \rightarrow 0} \frac{f(c + h) - f(c)}{h} = \lim_{h \rightarrow 0^-} \frac{f(c + h) - f(c)}{h} \geq 0$$

We have shown that $f'(c) \geq 0$ and also that $f'(c) \leq 0$. Since both of these inequalities must be true, the only possibility is that $f'(c) = 0$.

We have proved Fermat's Theorem for the case of a local maximum. The case of a local minimum can be proved in a similar manner. ■

This theorem was stated and used in Section 4.5. Part (b) has a similar proof.

(3) The Stability Criterion for Recursive Sequences Suppose that $\hat{x}$ is an equilibrium of the recursive sequence $x_{t+1} = f(x_t)$, where f' is continuous.

- (a) If $|f'(\hat{x})| < 1$, the equilibrium is stable.
- (b) If $|f'(\hat{x})| > 1$, the equilibrium is unstable.

PROOF OF PART (A) We want to show that if the sequence $\{x_t\}$ starts with x_0 sufficiently close to the equilibrium $\hat{x}$, then $x_t \rightarrow \hat{x}$ as $t \rightarrow \infty$.

By the Mean Value Theorem there is a number c between x_0 and $\hat{x}$ such that

$$f(x_0) - f(\hat{x}) = f'(c)(x_0 - \hat{x})$$

But $f(\hat{x}) = \hat{x}$, so we can write

$$x_1 - \hat{x} = f'(c)(x_0 - \hat{x})$$

Because f' is continuous and $|f'(\hat{x})| < 1$, we can guarantee that $|f'(x)| < 1$ when x is sufficiently close to $\hat{x}$. In fact there is a positive number α such that

$$|f'(x)| \leq \alpha < 1 \quad \text{when } x \text{ is sufficiently close to } \hat{x}, \text{ say } |x - \hat{x}| < \delta$$

So if $|x_0 - \hat{x}| < \delta$, then

$$|x_1 - \hat{x}| = |f'(c)||x_0 - \hat{x}| \leq \alpha|x_0 - \hat{x}|$$

Using this argument t times, we get

$$|x_t - \hat{x}| \leq \alpha^t |x_0 - \hat{x}|$$

Because $0 < \alpha < 1$, we know that $\alpha^t \rightarrow 0$ as $t \rightarrow \infty$. Therefore $|x_t - \hat{x}| \rightarrow 0$ as $t \rightarrow \infty$. This means that $x_t \rightarrow \hat{x}$ as $t \rightarrow \infty$ and so $\hat{x}$ is a stable equilibrium. ■

Clairaut's Theorem was discussed in Section 9.2.

Clairaut's Theorem Suppose f is defined on a disk D that contains the point (a, b) . If the functions f_{xy} and f_{yx} are both continuous on D , then

$$f_{xy}(a, b) = f_{yx}(a, b)$$

PROOF For small values of h , $h \neq 0$, consider the difference

$$\Delta(h) = [f(a + h, b + h) - f(a + h, b)] - [f(a, b + h) - f(a, b)]$$

Notice that if we let $g(x) = f(x, b + h) - f(x, b)$, then

$$\Delta(h) = g(a + h) - g(a)$$

By the Mean Value Theorem, there is a number c between a and $a + h$ such that

$$g(a + h) - g(a) = g'(c)h = h[f_x(c, b + h) - f_x(c, b)]$$

Applying the Mean Value Theorem again, this time to f_x , we get a number d between b and $b + h$ such that

$$f_x(c, b + h) - f_x(c, b) = f_{xy}(c, d)h$$

Combining these equations, we obtain

$$\Delta(h) = h^2 f_{xy}(c, d)$$

If $h \rightarrow 0$, then $(c, d) \rightarrow (a, b)$, so the continuity of f_{xy} at (a, b) gives

$$\lim_{h \rightarrow 0} \frac{\Delta(h)}{h^2} = \lim_{(c, d) \rightarrow (a, b)} f_{xy}(c, d) = f_{xy}(a, b)$$

Similarly, by writing

$$\Delta(h) = [f(a + h, b + h) - f(a, b + h)] - [f(a + h, b) - f(a, b)]$$

and using the Mean Value Theorem twice and the continuity of f_{yx} at (a, b) , we obtain

$$\lim_{h \rightarrow 0} \frac{\Delta(h)}{h^2} = f_{yx}(a, b)$$

It follows that $f_{xy}(a, b) = f_{yx}(a, b)$. ■

The Second Derivatives Test was discussed in Section 9.6. Parts (b) and (c) have similar proofs.

Second Derivatives Test Suppose the second partial derivatives of f are continuous on a disk with center (a, b) , and suppose that $f_x(a, b) = 0$ and $f_y(a, b) = 0$ [that is, (a, b) is a critical point of f]. Let

$$D = D(a, b) = f_{xx}(a, b)f_{yy}(a, b) - [f_{xy}(a, b)]^2$$

- (a) If $D > 0$ and $f_{xx}(a, b) > 0$, then $f(a, b)$ is a local minimum.
- (b) If $D > 0$ and $f_{xx}(a, b) < 0$, then $f(a, b)$ is a local maximum.
- (c) If $D < 0$, then $f(a, b)$ is not a local maximum or minimum.

PROOF OF PART (A) We compute the second-order directional derivative of f in the direction of $\mathbf{u} = [h, k]$. The first-order derivative is given by Theorem 9.5.3:

$$D_{\mathbf{u}}f = f_x h + f_y k$$

Applying this theorem a second time, we have

$$\begin{aligned} D_{\mathbf{u}}^2 f &= D_{\mathbf{u}}(D_{\mathbf{u}}f) = \frac{\partial}{\partial x}(D_{\mathbf{u}}f)h + \frac{\partial}{\partial y}(D_{\mathbf{u}}f)k \\ &= (f_{xx}h + f_{yx}k)h + (f_{xy}h + f_{yy}k)k \\ &= f_{xx}h^2 + 2f_{xy}hk + f_{yy}k^2 \end{aligned} \quad \text{(by Clairaut's Theorem)}$$

If we complete the square in this expression, we obtain

$$(4) \quad D_{\mathbf{u}}^2 f = f_{xx} \left(h + \frac{f_{xy}}{f_{xx}} k \right)^2 + \frac{k^2}{f_{xx}} (f_{xx}f_{yy} - f_{xy}^2)$$

We are given that $f_{xx}(a, b) > 0$ and $D(a, b) > 0$. But f_{xx} and $D = f_{xx}f_{yy} - f_{xy}^2$ are continuous functions, so there is a disk B with center (a, b) and radius $\delta > 0$ such that $f_{xx}(x, y) > 0$ and $D(x, y) > 0$ whenever (x, y) is in B . Therefore, by looking at Equation 4, we see that $D_{\mathbf{u}}^2 f(x, y) > 0$ whenever (x, y) is in B . This means that if C is the curve obtained by intersecting the graph of f with the vertical plane through $P(a, b, f(a, b))$ in the direction of $\mathbf{u}$, then C is concave upward on an interval of length 2δ . This is true in the direction of every vector $\mathbf{u}$, so if we restrict (x, y) to lie in B , the graph of f lies above its horizontal tangent plane at P . Thus $f(x, y) \geq f(a, b)$ whenever (x, y) is in B . This shows that $f(a, b)$ is a local minimum. ■

F Sigma Notation

A convenient way of writing sums uses the Greek letter Σ (capital sigma, corresponding to our letter S) and is called **sigma notation**.

This tells us to end with $i = n$.
This tells us to add.
This tells us to start with $i = m$.

$$\sum_{i=m}^n a_i$$

(1) Definition If $a_m, a_{m+1}, \dots, a_n$ are real numbers and m and n are integers such that $m \leq n$, then

$$\sum_{i=m}^n a_i = a_m + a_{m+1} + a_{m+2} + \cdots + a_{n-1} + a_n$$

With function notation, Definition 1 can be written as

$$\sum_{i=m}^n f(i) = f(m) + f(m+1) + f(m+2) + \cdots + f(n-1) + f(n)$$

Thus the symbol $\sum_{i=m}^n$ indicates a summation in which the letter i (called the **index of summation**) takes on consecutive integer values beginning with m and ending with n , that is, $m, m+1, \dots, n$. Other letters can also be used as the index of summation.

EXAMPLE 1

$$(a) \sum_{i=1}^4 i^2 = 1^2 + 2^2 + 3^2 + 4^2 = 30$$

$$(b) \sum_{i=3}^n i = 3 + 4 + 5 + \cdots + (n-1) + n$$

$$(c) \sum_{j=0}^5 2^j = 2^0 + 2^1 + 2^2 + 2^3 + 2^4 + 2^5 = 63$$

$$(d) \sum_{k=1}^n \frac{1}{k} = 1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{n}$$

$$(e) \sum_{i=1}^3 \frac{i-1}{i^2+3} = \frac{1-1}{1^2+3} + \frac{2-1}{2^2+3} + \frac{3-1}{3^2+3} = 0 + \frac{1}{7} + \frac{1}{6} = \frac{13}{42}$$

$$(f) \sum_{i=1}^4 2 = 2 + 2 + 2 + 2 = 8$$

EXAMPLE 2 | Write the sum $2^3 + 3^3 + \cdots + n^3$ in sigma notation.

SOLUTION There is no unique way of writing a sum in sigma notation. We could write

$$2^3 + 3^3 + \cdots + n^3 = \sum_{i=2}^n i^3$$

$$\text{or} \quad 2^3 + 3^3 + \cdots + n^3 = \sum_{j=1}^{n-1} (j+1)^3$$

$$\text{or} \quad 2^3 + 3^3 + \cdots + n^3 = \sum_{k=0}^{n-2} (k+2)^3$$

The following theorem gives three simple rules for working with sigma notation.

(2) Theorem If c is any constant (that is, it does not depend on i), then

$$(a) \sum_{i=m}^n ca_i = c \sum_{i=m}^n a_i \quad (b) \sum_{i=m}^n (a_i + b_i) = \sum_{i=m}^n a_i + \sum_{i=m}^n b_i$$

$$(c) \sum_{i=m}^n (a_i - b_i) = \sum_{i=m}^n a_i - \sum_{i=m}^n b_i$$

PROOF To see why these rules are true, all we have to do is write both sides in expanded form. Rule (a) is just the distributive property of real numbers:

$$ca_m + ca_{m+1} + \cdots + ca_n = c(a_m + a_{m+1} + \cdots + a_n)$$

Rule (b) follows from the associative and commutative properties:

$$\begin{aligned}(a_m + b_m) + (a_{m+1} + b_{m+1}) + \cdots + (a_n + b_n) \\ = (a_m + a_{m+1} + \cdots + a_n) + (b_m + b_{m+1} + \cdots + b_n)\end{aligned}$$

Rule (c) is proved similarly. ■

EXAMPLE 3 | Find $\sum_{i=1}^n 1$.

SOLUTION $\sum_{i=1}^n 1 = \underbrace{1 + 1 + \cdots + 1}_{n \text{ terms}} = n$ ■

EXAMPLE 4 | Prove the formula for the sum of the first n positive integers:

$$\sum_{i=1}^n i = 1 + 2 + 3 + \cdots + n = \frac{n(n+1)}{2}$$

SOLUTION This formula can be proved by mathematical induction or by the following method used by the German mathematician Karl Friedrich Gauss (1777–1855) when he was ten years old.

Write the sum S twice, once in the usual order and once in reverse order:

$$\begin{aligned}S &= 1 + 2 + 3 + \cdots + (n-1) + n \\ S &= n + (n-1) + (n-2) + \cdots + 2 + 1\end{aligned}$$

Adding all columns vertically, we get

$$2S = (n+1) + (n+1) + (n+1) + \cdots + (n+1) + (n+1)$$

On the right side there are n terms, each of which is $n+1$, so

$$2S = n(n+1) \quad \text{or} \quad S = \frac{n(n+1)}{2}$$
 ■

EXAMPLE 5 | Prove the formula for the sum of the squares of the first n positive integers:

$$\sum_{i=1}^n i^2 = 1^2 + 2^2 + 3^2 + \cdots + n^2 = \frac{n(n+1)(2n+1)}{6}$$

SOLUTION 1 Let S be the desired sum. We start with the *telescoping sum* (or collapsing sum):

Most terms cancel in pairs.

$$\begin{aligned}\sum_{i=1}^n [(1+i)^3 - i^3] &= (\cancel{2^3} - 1^3) + (\cancel{3^3} - \cancel{2^3}) + (\cancel{4^3} - \cancel{3^3}) + \cdots + [(n+1)^3 - \cancel{n^3}] \\ &= (n+1)^3 - 1^3 = n^3 + 3n^2 + 3n\end{aligned}$$

On the other hand, using Theorem 2 and Examples 3 and 4, we have

$$\begin{aligned}\sum_{i=1}^n [(1+i)^3 - i^3] &= \sum_{i=1}^n [3i^2 + 3i + 1] = 3 \sum_{i=1}^n i^2 + 3 \sum_{i=1}^n i + \sum_{i=1}^n 1 \\ &= 3S + 3 \frac{n(n+1)}{2} + n = 3S + \frac{3}{2}n^2 + \frac{5}{2}n\end{aligned}$$

Thus we have

$$n^3 + 3n^2 + 3n = 3S + \frac{3}{2}n^2 + \frac{5}{2}n$$

Solving this equation for S , we obtain

$$3S = n^3 + \frac{3}{2}n^2 + \frac{1}{2}n$$

or
$$S = \frac{2n^3 + 3n^2 + n}{6} = \frac{n(n+1)(2n+1)}{6}$$

Principle of Mathematical Induction

Let S_n be a statement involving the positive integer n . Suppose that

1. S_1 is true.
2. If S_k is true, then S_{k+1} is true.

Then S_n is true for all positive integers n .

SOLUTION 2 Let S_n be the given formula.

1. S_1 is true because $1^2 = \frac{1(1+1)(2 \cdot 1 + 1)}{6}$
2. Assume that S_k is true; that is,

$$1^2 + 2^2 + 3^2 + \cdots + k^2 = \frac{k(k+1)(2k+1)}{6}$$

Then

$$\begin{aligned} 1^2 + 2^2 + 3^2 + \cdots + (k+1)^2 &= (1^2 + 2^2 + 3^2 + \cdots + k^2) + (k+1)^2 \\ &= \frac{k(k+1)(2k+1)}{6} + (k+1)^2 \\ &= (k+1) \frac{k(2k+1) + 6(k+1)}{6} \\ &= (k+1) \frac{2k^2 + 7k + 6}{6} \\ &= \frac{(k+1)(k+2)(2k+3)}{6} \\ &= \frac{(k+1)[(k+1)+1][2(k+1)+1]}{6} \end{aligned}$$

So S_{k+1} is true.

By the Principle of Mathematical Induction, S_n is true for all n . ■

We list the results of Examples 3, 4, and 5 together with a similar result for cubes (see Exercises 37–40) as Theorem 3. These formulas are needed for finding areas and evaluating integrals in Chapter 5.

(3) Theorem Let c be a constant and n a positive integer. Then

(a) $\sum_{i=1}^n 1 = n$

(b) $\sum_{i=1}^n c = nc$

(c) $\sum_{i=1}^n i = \frac{n(n+1)}{2}$

(d) $\sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{6}$

(e) $\sum_{i=1}^n i^3 = \left[\frac{n(n+1)}{2} \right]^2$

EXAMPLE 6 | Evaluate $\sum_{i=1}^n i(4i^2 - 3)$.

SOLUTION Using Theorems 2 and 3, we have

$$\begin{aligned}\sum_{i=1}^n i(4i^2 - 3) &= \sum_{i=1}^n (4i^3 - 3i) = 4 \sum_{i=1}^n i^3 - 3 \sum_{i=1}^n i \\ &= 4 \left[\frac{n(n+1)}{2} \right]^2 - 3 \frac{n(n+1)}{2} \\ &= \frac{n(n+1)[2n(n+1) - 3]}{2} \\ &= \frac{n(n+1)(2n^2 + 2n - 3)}{2}\end{aligned}$$

The type of calculation in Example 7 arises in Chapter 5 when we compute areas.

EXAMPLE 7 | Find $\lim_{n \rightarrow \infty} \sum_{i=1}^n \frac{3}{n} \left[\left(\frac{i}{n} \right)^2 + 1 \right]$.

SOLUTION

$$\begin{aligned}\lim_{n \rightarrow \infty} \sum_{i=1}^n \frac{3}{n} \left[\left(\frac{i}{n} \right)^2 + 1 \right] &= \lim_{n \rightarrow \infty} \sum_{i=1}^n \left[\frac{3}{n^3} i^2 + \frac{3}{n} \right] \\ &= \lim_{n \rightarrow \infty} \left[\frac{3}{n^3} \sum_{i=1}^n i^2 + \frac{3}{n} \sum_{i=1}^n 1 \right] \\ &= \lim_{n \rightarrow \infty} \left[\frac{3}{n^3} \frac{n(n+1)(2n+1)}{6} + \frac{3}{n} \cdot n \right] \\ &= \lim_{n \rightarrow \infty} \left[\frac{1}{2} \cdot \frac{n}{n} \cdot \left(\frac{n+1}{n} \right) \left(\frac{2n+1}{n} \right) + 3 \right] \\ &= \lim_{n \rightarrow \infty} \left[\frac{1}{2} \cdot 1 \left(1 + \frac{1}{n} \right) \left(2 + \frac{1}{n} \right) + 3 \right] \\ &= \frac{1}{2} \cdot 1 \cdot 1 \cdot 2 + 3 = 4\end{aligned}$$

EXERCISES F

1–10 Write the sum in expanded form.

1. $\sum_{i=1}^5 \sqrt{i}$

2. $\sum_{i=1}^6 \frac{1}{i+1}$

3. $\sum_{i=4}^6 3^i$

4. $\sum_{i=4}^6 i^3$

5. $\sum_{k=0}^4 \frac{2k-1}{2k+1}$

6. $\sum_{k=5}^8 x^k$

7. $\sum_{i=1}^n i^{10}$

8. $\sum_{j=n}^{n+3} j^2$

9. $\sum_{j=0}^{n-1} (-1)^j$

10. $\sum_{i=1}^n f(x_i) \Delta x_i$

11–20 Write the sum in sigma notation.

11. $1 + 2 + 3 + 4 + \cdots + 10$

12. $\sqrt{3} + \sqrt{4} + \sqrt{5} + \sqrt{6} + \sqrt{7}$
13. $\frac{1}{2} + \frac{2}{3} + \frac{3}{4} + \frac{4}{5} + \cdots + \frac{19}{20}$
14. $\frac{3}{7} + \frac{4}{8} + \frac{5}{9} + \frac{6}{10} + \cdots + \frac{23}{27}$
15. $2 + 4 + 6 + 8 + \cdots + 2n$
16. $1 + 3 + 5 + 7 + \cdots + (2n - 1)$
17. $1 + 2 + 4 + 8 + 16 + 32$
18. $\frac{1}{1} + \frac{1}{4} + \frac{1}{9} + \frac{1}{16} + \frac{1}{25} + \frac{1}{36}$
19. $x + x^2 + x^3 + \cdots + x^n$
20. $1 - x + x^2 - x^3 + \cdots + (-1)^n x^n$

21–35 Find the value of the sum.

21. $\sum_{i=4}^8 (3i - 2)$
22. $\sum_{i=3}^6 i(i + 2)$
23. $\sum_{j=1}^6 3^{j+1}$
24. $\sum_{k=0}^8 \cos k\pi$
25. $\sum_{n=1}^{20} (-1)^n$
26. $\sum_{i=1}^{100} 4$
27. $\sum_{i=0}^4 (2^i + i^2)$
28. $\sum_{i=-2}^4 2^{3-i}$
29. $\sum_{i=1}^n 2i$
30. $\sum_{i=1}^n (2 - 5i)$
31. $\sum_{i=1}^n (i^2 + 3i + 4)$
32. $\sum_{i=1}^n (3 + 2i)^2$
33. $\sum_{i=1}^n (i + 1)(i + 2)$
34. $\sum_{i=1}^n i(i + 1)(i + 2)$
35. $\sum_{i=1}^n (i^3 - i - 2)$

36. Find the number n such that $\sum_{i=1}^n i = 78$.

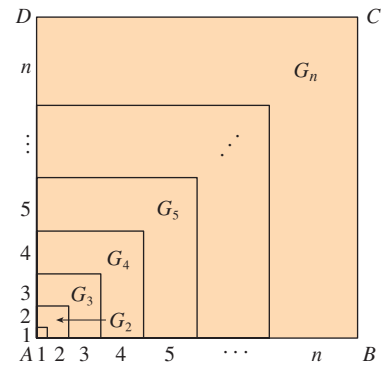
37. Prove formula (b) of Theorem 3.

38. Prove formula (e) of Theorem 3 using mathematical induction.

39. Prove formula (e) of Theorem 3 using a method similar to that of Example 5, Solution 1 [start with $(1 + i)^4 - i^4$].

40. Prove formula (e) of Theorem 3 using the following method published by Abu Bekr Mohammed ibn Alhusain Alkarchi in about AD 1010. The figure shows a square $ABCD$ in which sides AB and AD have been divided into segments of lengths $1, 2, 3, \dots, n$. Thus the side of the square has length $n(n + 1)/2$ so the area is $[n(n + 1)/2]^2$. But the area is also

the sum of the areas of the n “gnomons” $G_1, G_2, \dots, G_n$ shown in the figure. Show that the area of G_i is i^3 and conclude that formula (e) is true.



41. Evaluate each telescoping sum.

- (a) $\sum_{i=1}^n [i^4 - (i - 1)^4]$
- (b) $\sum_{i=1}^{100} (5^i - 5^{i-1})$
- (c) $\sum_{i=3}^{99} \left(\frac{1}{i} - \frac{1}{i + 1} \right)$
- (d) $\sum_{i=1}^n (a_i - a_{i-1})$

42. Prove the generalized triangle inequality:

$$\left| \sum_{i=1}^n a_i \right| \leq \sum_{i=1}^n |a_i|$$

43–46 Find the limit.

43. $\lim_{n \rightarrow \infty} \sum_{i=1}^n \frac{1}{n} \left(\frac{i}{n} \right)^2$
44. $\lim_{n \rightarrow \infty} \sum_{i=1}^n \frac{1}{n} \left[\left(\frac{i}{n} \right)^3 + 1 \right]$
45. $\lim_{n \rightarrow \infty} \sum_{i=1}^n \frac{2}{n} \left[\left(\frac{2i}{n} \right)^3 + 5 \left(\frac{2i}{n} \right) \right]$
46. $\lim_{n \rightarrow \infty} \sum_{i=1}^n \frac{3}{n} \left[\left(1 + \frac{3i}{n} \right)^3 - 2 \left(1 + \frac{3i}{n} \right) \right]$

47. Prove the formula for the sum of a finite geometric series with first term a and common ratio $r \neq 1$:

$$\sum_{i=1}^n ar^{i-1} = a + ar + ar^2 + \cdots + ar^{n-1} = \frac{a(r^n - 1)}{r - 1}$$

48. Evaluate $\sum_{i=1}^n \frac{3}{2^{i-1}}$.

49. Evaluate $\sum_{i=1}^n (2i + 2^i)$.

50. Evaluate $\sum_{i=1}^m \left[\sum_{j=1}^n (i + j) \right]$.

G Complex Numbers

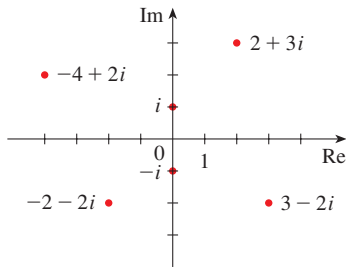


FIGURE 1
Complex numbers as points in the Argand plane

A **complex number** can be represented by an expression of the form $a + bi$, where a and b are real numbers and i is a symbol with the property that $i^2 = -1$. The complex number $a + bi$ can also be represented by the ordered pair (a, b) and plotted as a point in a plane (called the Argand plane) as in Figure 1. Thus the complex number $i = 0 + 1 \cdot i$ is identified with the point $(0, 1)$.

The **real part** of the complex number $a + bi$ is the real number a and the **imaginary part** is the real number b . Thus the real part of $4 - 3i$ is 4 and the imaginary part is -3 . Two complex numbers $a + bi$ and $c + di$ are **equal** if $a = c$ and $b = d$, that is, their real parts are equal and their imaginary parts are equal. In the Argand plane the horizontal axis is called the real axis and the vertical axis is called the imaginary axis.

The sum and difference of two complex numbers are defined by adding or subtracting their real parts and their imaginary parts:

$$(a + bi) + (c + di) = (a + c) + (b + d)i$$

$$(a + bi) - (c + di) = (a - c) + (b - d)i$$

For instance,

$$(1 - i) + (4 + 7i) = (1 + 4) + (-1 + 7)i = 5 + 6i$$

The product of complex numbers is defined so that the usual commutative and distributive laws hold:

$$\begin{aligned}(a + bi)(c + di) &= a(c + di) + (bi)(c + di) \\ &= ac + adi + bci + bdi^2\end{aligned}$$

Since $i^2 = -1$, this becomes

$$(a + bi)(c + di) = (ac - bd) + (ad + bc)i$$

EXAMPLE 1

$$\begin{aligned}(-1 + 3i)(2 - 5i) &= (-1)(2 - 5i) + 3i(2 - 5i) \\ &= -2 + 5i + 6i - 15(-1) = 13 + 11i\end{aligned}$$

Division of complex numbers is much like rationalizing the denominator of a rational expression. For the complex number $z = a + bi$, we define its **complex conjugate** to be $\bar{z} = a - bi$. To find the quotient of two complex numbers we multiply numerator and denominator by the complex conjugate of the denominator.

EXAMPLE 2 | Express the number $\frac{-1 + 3i}{2 + 5i}$ in the form $a + bi$.

SOLUTION We multiply numerator and denominator by the complex conjugate of $2 + 5i$, namely, $2 - 5i$, and we take advantage of the result of Example 1:

$$\frac{-1 + 3i}{2 + 5i} = \frac{-1 + 3i}{2 + 5i} \cdot \frac{2 - 5i}{2 - 5i} = \frac{13 + 11i}{2^2 + 5^2} = \frac{13}{29} + \frac{11}{29}i$$

The geometric interpretation of the complex conjugate is shown in Figure 2: $\bar{z}$ is the reflection of z in the real axis. We list some of the properties of the complex conjugate

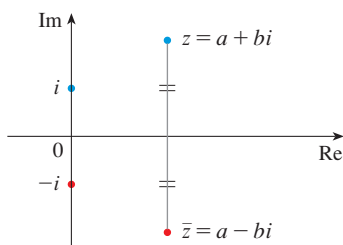


FIGURE 2

in the following box. The proofs follow from the definition and are requested in Exercise 18.

Properties of Conjugates

$$\overline{z + w} = \bar{z} + \bar{w} \quad \overline{zw} = \bar{z} \bar{w} \quad \overline{z^n} = \bar{z}^n$$

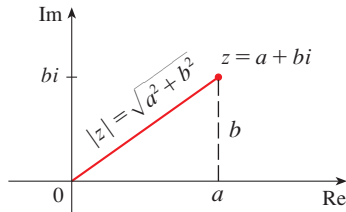


FIGURE 3

The **modulus**, or **absolute value**, $|z|$ of a complex number $z = a + bi$ is its distance from the origin. From Figure 3 we see that if $z = a + bi$, then

$$|z| = \sqrt{a^2 + b^2}$$

Notice that

$$z\bar{z} = (a + bi)(a - bi) = a^2 + abi - abi - b^2i^2 = a^2 + b^2$$

and so

$$z\bar{z} = |z|^2$$

This explains why the division procedure in Example 2 works in general:

$$\frac{z}{w} = \frac{z\bar{w}}{w\bar{w}} = \frac{z\bar{w}}{|w|^2}$$

Since $i^2 = -1$, we can think of i as a square root of -1 . But notice that we also have $(-i)^2 = i^2 = -1$ and so $-i$ is also a square root of -1 . We say that i is the **principal square root** of -1 and write $\sqrt{-1} = i$. In general, if c is any positive number, we write

$$\sqrt{-c} = \sqrt{c} i$$

With this convention, the usual derivation and formula for the roots of the quadratic equation $ax^2 + bx + c = 0$ are valid even when $b^2 - 4ac < 0$:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

EXAMPLE 3 | Find the roots of the equation $x^2 + x + 1 = 0$.

SOLUTION Using the quadratic formula, we have

$$x = \frac{-1 \pm \sqrt{1^2 - 4 \cdot 1}}{2} = \frac{-1 \pm \sqrt{-3}}{2} = \frac{-1 \pm \sqrt{3} i}{2}$$

We observe that the solutions of the equation in Example 3 are complex conjugates of each other. In general, the solutions of any quadratic equation $ax^2 + bx + c = 0$ with real coefficients a , b , and c are always complex conjugates. (If z is real, $\bar{z} = z$, so z is its own conjugate.)

We have seen that if we allow complex numbers as solutions, then every quadratic equation has a solution. More generally, it is true that every polynomial equation

$$a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0 = 0$$

of degree at least one has a solution among the complex numbers. This fact is known as the Fundamental Theorem of Algebra and was proved by Gauss.

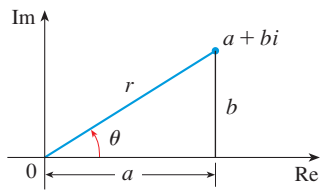


FIGURE 4

■ Polar Form

Any complex number $z = a + bi$ can be considered as a point (a, b) and any such point can be represented by **polar coordinates** (r, θ) with $r \geq 0$. In fact,

$$a = r \cos \theta \quad b = r \sin \theta$$

as in Figure 4. Therefore we have

$$z = a + bi = (r \cos \theta) + (r \sin \theta)i$$

Thus we can write any complex number z in the form

$$z = r(\cos \theta + i \sin \theta)$$

where $r = |z| = \sqrt{a^2 + b^2}$ and $\tan \theta = \frac{b}{a}$

The angle θ is called the **argument** of z and we write $\theta = \arg(z)$. Note that $\arg(z)$ is not unique; any two arguments of z differ by an integer multiple of 2π .

EXAMPLE 4 | Write the following numbers in polar form.

(a) $z = 1 + i$

(b) $w = \sqrt{3} - i$

SOLUTION

(a) We have $r = |z| = \sqrt{1^2 + 1^2} = \sqrt{2}$ and $\tan \theta = 1$, so we can take $\theta = \pi/4$. Therefore the polar form is

$$z = \sqrt{2} \left(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4} \right)$$

(b) Here we have $r = |w| = \sqrt{3 + 1} = 2$ and $\tan \theta = -1/\sqrt{3}$. Since w lies in the fourth quadrant, we take $\theta = -\pi/6$ and

$$w = 2 \left[\cos \left(-\frac{\pi}{6} \right) + i \sin \left(-\frac{\pi}{6} \right) \right]$$

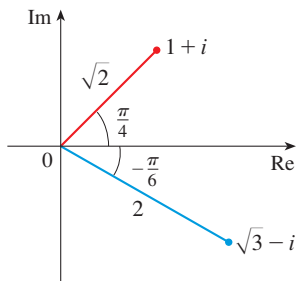


FIGURE 5

The numbers z and w are shown in Figure 5. ■

The polar form of complex numbers gives insight into multiplication and division. Let

$$z_1 = r_1(\cos \theta_1 + i \sin \theta_1) \quad z_2 = r_2(\cos \theta_2 + i \sin \theta_2)$$

be two complex numbers written in polar form. Then

$$\begin{aligned} z_1 z_2 &= r_1 r_2 (\cos \theta_1 + i \sin \theta_1)(\cos \theta_2 + i \sin \theta_2) \\ &= r_1 r_2 [(\cos \theta_1 \cos \theta_2 - \sin \theta_1 \sin \theta_2) + i(\sin \theta_1 \cos \theta_2 + \cos \theta_1 \sin \theta_2)] \end{aligned}$$

Therefore, using the addition formulas for cosine and sine, we have

(1)

$$z_1 z_2 = r_1 r_2 [\cos(\theta_1 + \theta_2) + i \sin(\theta_1 + \theta_2)]$$

This formula says that to *multiply two complex numbers we multiply the moduli and add the arguments*. (See Figure 6.)

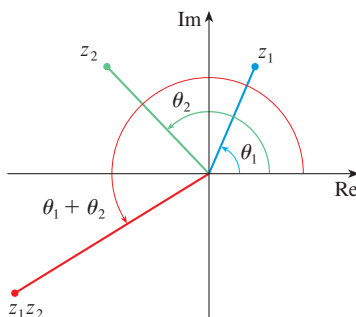


FIGURE 6

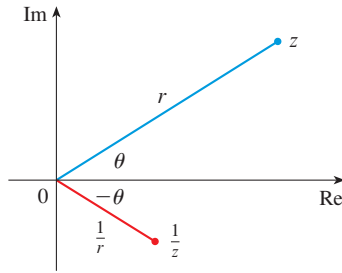


FIGURE 7

A similar argument using the subtraction formulas for sine and cosine shows that to divide two complex numbers we divide the moduli and subtract the arguments.

$$\frac{z_1}{z_2} = \frac{r_1}{r_2} [\cos(\theta_1 - \theta_2) + i \sin(\theta_1 - \theta_2)] \quad z_2 \neq 0$$

In particular, taking $z_1 = 1$ and $z_2 = z$ (and therefore $\theta_1 = 0$ and $\theta_2 = \theta$), we have the following, which is illustrated in Figure 7.

$$\text{If } z = r(\cos \theta + i \sin \theta), \text{ then } \frac{1}{z} = \frac{1}{r}(\cos \theta - i \sin \theta).$$

EXAMPLE 5 | Find the product of the complex numbers $1 + i$ and $\sqrt{3} - i$ in polar form.

SOLUTION From Example 4 we have

$$1 + i = \sqrt{2} \left(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4} \right)$$

and

$$\sqrt{3} - i = 2 \left[\cos \left(-\frac{\pi}{6} \right) + i \sin \left(-\frac{\pi}{6} \right) \right]$$

So, by Equation 1,

$$\begin{aligned} (1 + i)(\sqrt{3} - i) &= 2\sqrt{2} \left[\cos \left(\frac{\pi}{4} - \frac{\pi}{6} \right) + i \sin \left(\frac{\pi}{4} - \frac{\pi}{6} \right) \right] \\ &= 2\sqrt{2} \left(\cos \frac{\pi}{12} + i \sin \frac{\pi}{12} \right) \end{aligned}$$

This is illustrated in Figure 8. ■

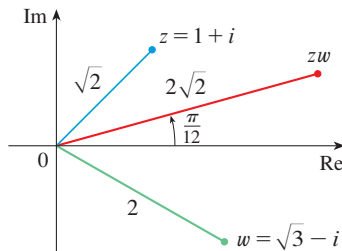


FIGURE 8

Repeated use of Formula 1 shows how to compute powers of a complex number. If

$$z = r(\cos \theta + i \sin \theta)$$

then

$$z^2 = r^2(\cos 2\theta + i \sin 2\theta)$$

and

$$z^3 = zz^2 = r^3(\cos 3\theta + i \sin 3\theta)$$

In general, we obtain the following result, which is named after the French mathematician Abraham De Moivre (1667–1754).

(2) De Moivre's Theorem If $z = r(\cos \theta + i \sin \theta)$ and n is a positive integer, then

$$z^n = [r(\cos \theta + i \sin \theta)]^n = r^n(\cos n\theta + i \sin n\theta)$$

This says that *to take the n th power of a complex number we take the n th power of the modulus and multiply the argument by n .*

EXAMPLE 6 | Find $(\frac{1}{2} + \frac{1}{2}i)^{10}$.

SOLUTION Since $\frac{1}{2} + \frac{1}{2}i = \frac{1}{2}(1 + i)$, it follows from Example 4(a) that $\frac{1}{2} + \frac{1}{2}i$ has the polar form

$$\frac{1}{2} + \frac{1}{2}i = \frac{\sqrt{2}}{2} \left(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4} \right)$$

So by De Moivre's Theorem,

$$\begin{aligned} \left(\frac{1}{2} + \frac{1}{2}i \right)^{10} &= \left(\frac{\sqrt{2}}{2} \right)^{10} \left(\cos \frac{10\pi}{4} + i \sin \frac{10\pi}{4} \right) \\ &= \frac{2^5}{2^{10}} \left(\cos \frac{5\pi}{2} + i \sin \frac{5\pi}{2} \right) = \frac{1}{32} i \end{aligned}$$

De Moivre's Theorem can also be used to find the n th roots of complex numbers. An n th root of the complex number z is a complex number w such that

$$w^n = z$$

Writing these two numbers in polar form as

$$w = s(\cos \phi + i \sin \phi) \quad \text{and} \quad z = r(\cos \theta + i \sin \theta)$$

and using De Moivre's Theorem, we get

$$s^n(\cos n\phi + i \sin n\phi) = r(\cos \theta + i \sin \theta)$$

The equality of these two complex numbers shows that

$$s^n = r \quad \text{or} \quad s = r^{1/n}$$

$$\text{and} \quad \cos n\phi = \cos \theta \quad \text{and} \quad \sin n\phi = \sin \theta$$

From the fact that sine and cosine have period 2π , it follows that

$$n\phi = \theta + 2k\pi \quad \text{or} \quad \phi = \frac{\theta + 2k\pi}{n}$$

Thus

$$w = r^{1/n} \left[\cos \left(\frac{\theta + 2k\pi}{n} \right) + i \sin \left(\frac{\theta + 2k\pi}{n} \right) \right]$$

Since this expression gives a different value of w for $k = 0, 1, 2, \dots, n-1$, we have the following.

(3) Roots of a Complex Number Let $z = r(\cos \theta + i \sin \theta)$ and let n be a positive integer. Then z has the n distinct n th roots

$$w_k = r^{1/n} \left[\cos \left(\frac{\theta + 2k\pi}{n} \right) + i \sin \left(\frac{\theta + 2k\pi}{n} \right) \right]$$

where $k = 0, 1, 2, \dots, n-1$.

Notice that each of the n th roots of z has modulus $|w_k| = r^{1/n}$. Thus all the n th roots of z lie on the circle of radius $r^{1/n}$ in the complex plane. Also, since the argument of each suc-

cessive n th root exceeds the argument of the previous root by $2\pi/n$, we see that the n th roots of z are equally spaced on this circle.

EXAMPLE 7 | Find the six sixth roots of $z = -8$ and graph these roots in the complex plane.

SOLUTION In polar form, $z = 8(\cos \pi + i \sin \pi)$. Applying Equation 3 with $n = 6$, we get

$$w_k = 8^{1/6} \left(\cos \frac{\pi + 2k\pi}{6} + i \sin \frac{\pi + 2k\pi}{6} \right)$$

We get the six sixth roots of -8 by taking $k = 0, 1, 2, 3, 4, 5$ in this formula:

$$w_0 = 8^{1/6} \left(\cos \frac{\pi}{6} + i \sin \frac{\pi}{6} \right) = \sqrt{2} \left(\frac{\sqrt{3}}{2} + \frac{1}{2} i \right)$$

$$w_1 = 8^{1/6} \left(\cos \frac{\pi}{2} + i \sin \frac{\pi}{2} \right) = \sqrt{2} i$$

$$w_2 = 8^{1/6} \left(\cos \frac{5\pi}{6} + i \sin \frac{5\pi}{6} \right) = \sqrt{2} \left(-\frac{\sqrt{3}}{2} + \frac{1}{2} i \right)$$

$$w_3 = 8^{1/6} \left(\cos \frac{7\pi}{6} + i \sin \frac{7\pi}{6} \right) = \sqrt{2} \left(-\frac{\sqrt{3}}{2} - \frac{1}{2} i \right)$$

$$w_4 = 8^{1/6} \left(\cos \frac{3\pi}{2} + i \sin \frac{3\pi}{2} \right) = -\sqrt{2} i$$

$$w_5 = 8^{1/6} \left(\cos \frac{11\pi}{6} + i \sin \frac{11\pi}{6} \right) = \sqrt{2} \left(\frac{\sqrt{3}}{2} - \frac{1}{2} i \right)$$

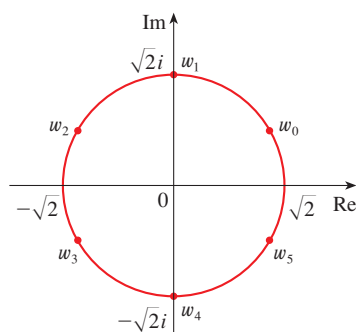


FIGURE 9
The six sixth roots of $z = -8$

All these points lie on the circle of radius $\sqrt{2}$ as shown in Figure 9. ■

Complex-Valued Functions and Their Derivatives

For a complex-valued function of one real variable t we can write

$$z(t) = x(t) + iy(t)$$

where $x(t)$ and $y(t)$ are the real and imaginary parts of $z(t)$. We can define its derivative $z'(t)$ by differentiating its real and imaginary parts:

$$z'(t) = x'(t) + iy'(t)$$

In particular, if we look at a complex number z on the unit circle in Figure 10 we can write it in polar form

$$z = \cos \theta + i \sin \theta$$

and consider the corresponding complex-valued function of the real variable θ

$$f(\theta) = \cos \theta + i \sin \theta$$

Then

$$f'(\theta) = \frac{d}{d\theta} (\cos \theta) + i \frac{d}{d\theta} (\sin \theta) = -\sin \theta + i \cos \theta$$

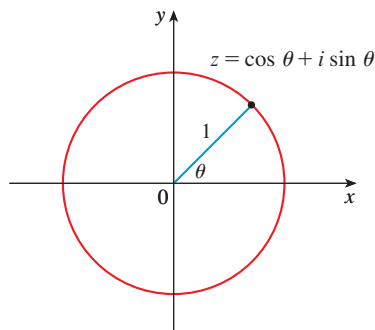


FIGURE 10

Because $i^2 = -1$, we could rewrite this as

$$\begin{aligned} f'(\theta) &= i \cos \theta + i^2 \sin \theta \\ &= i(\cos \theta + i \sin \theta) = if(\theta) \end{aligned}$$

The equation $f'(\theta) = if(\theta)$ shows that the derivative of f is proportional to the function itself, the proportionality constant being the complex number i . In Section 7.4 we showed that the only solution of the differential equation $dy/dt = ky$ is the exponential function $y(t) = Ae^{kt}$. Assuming that this is also true for complex functions, we get

$$f'(\theta) = if(\theta) \quad \Rightarrow \quad f(\theta) = Ae^{i\theta}$$

where A is a constant. Putting $\theta = 0$, we get $f(0) = Ae^0 = A$. But

$$f(0) = \cos 0 + i \sin 0 = 1$$

and so $A = 1$ and therefore $f(\theta) = e^{i\theta}$. We have thus arrived at a famous formula:

(4) Euler's Formula

$$e^{i\theta} = \cos \theta + i \sin \theta$$

We could write the result of Example 8 as

$$e^{i\pi} + 1 = 0$$

This equation relates the five most famous numbers in all of mathematics: 0, 1, e , i , and π .

EXAMPLE 8 | Evaluate $e^{i\pi}$.

SOLUTION From Euler's Formula (4) with $\theta = \pi$, we get

$$e^{i\pi} = \cos \pi + i \sin \pi = -1 + i(0) = -1$$

We could define the general complex exponential function e^z for any complex number $z = x + iy$ by writing

$$(5) \quad e^z = e^{x+iy} = e^x e^{iy} = e^x (\cos y + i \sin y)$$

EXAMPLE 9 | Evaluate $e^{-1+i\pi/2}$.

SOLUTION Using Equation 5 we get

$$e^{-1+i\pi/2} = e^{-1} \left(\cos \frac{\pi}{2} + i \sin \frac{\pi}{2} \right) = \frac{1}{e} [0 + i(1)] = \frac{i}{e}$$

Finally, we note that Euler's equation provides us with an easier method of proving De Moivre's Theorem:

$$[r(\cos \theta + i \sin \theta)]^n = (re^{i\theta})^n = r^n e^{in\theta} = r^n (\cos n\theta + i \sin n\theta)$$

EXERCISES G

1–14 Evaluate the expression and write your answer in the form $a + bi$.

- | | |
|----------------------------|--|
| 1. $(5 - 6i) + (3 + 2i)$ | 2. $(4 - \frac{1}{2}i) - (9 + \frac{5}{2}i)$ |
| 3. $(2 + 5i)(4 - i)$ | 4. $(1 - 2i)(8 - 3i)$ |
| 5. $\overline{12 + 7i}$ | 6. $\overline{2i(\frac{1}{2} - i)}$ |
| 7. $\frac{1 + 4i}{3 + 2i}$ | 8. $\frac{3 + 2i}{1 - 4i}$ |
| 9. $\frac{1}{1 + i}$ | 10. $\frac{3}{4 - 3i}$ |
| 11. i^3 | 12. i^{100} |
| 13. $\sqrt{-25}$ | 14. $\sqrt{-3}\sqrt{-12}$ |

15–17 Find the complex conjugate and the modulus of the number.

- | | |
|---------------|-----------------------|
| 15. $12 - 5i$ | 16. $-1 + 2\sqrt{2}i$ |
| 17. $-4i$ | |

18. Prove the following properties of complex numbers.

- (a) $\overline{z + w} = \overline{z} + \overline{w}$ (b) $\overline{zw} = \overline{z}\overline{w}$
 (c) $\overline{z^n} = \overline{z}^n$, where n is a positive integer
 [Hint: Write $z = a + bi$, $w = c + di$.]

19–24 Find all solutions of the equation.

- | | |
|------------------------|--|
| 19. $4x^2 + 9 = 0$ | 20. $x^4 = 1$ |
| 21. $x^2 + 2x + 5 = 0$ | 22. $2x^2 - 2x + 1 = 0$ |
| 23. $z^2 + z + 2 = 0$ | 24. $z^2 + \frac{1}{2}z + \frac{1}{4} = 0$ |

25–28 Write the number in polar form with argument between 0 and 2π .

- | | |
|---------------|---------------------|
| 25. $-3 + 3i$ | 26. $1 - \sqrt{3}i$ |
| 27. $3 + 4i$ | 28. $8i$ |

29–32 Find polar forms for zw , z/w , and $1/z$ by first putting z and w into polar form.

29. $z = \sqrt{3} + i$, $w = 1 + \sqrt{3}i$
 30. $z = 4\sqrt{3} - 4i$, $w = 8i$
 31. $z = 2\sqrt{3} - 2i$, $w = -1 + i$

32. $z = 4(\sqrt{3} + i)$, $w = -3 - 3i$

33–36 Find the indicated power using De Moivre's Theorem.

- | | |
|--------------------------|-------------------------|
| 33. $(1 + i)^{20}$ | 34. $(1 - \sqrt{3}i)^5$ |
| 35. $(2\sqrt{3} + 2i)^5$ | 36. $(1 - i)^8$ |

37–40 Find the indicated roots. Sketch the roots in the complex plane.

- | | |
|---------------------------|-------------------------------|
| 37. The eighth roots of 1 | 38. The fifth roots of 32 |
| 39. The cube roots of i | 40. The cube roots of $1 + i$ |

41–46 Write the number in the form $a + bi$.

- | | |
|------------------|------------------|
| 41. $e^{i\pi/2}$ | 42. $e^{2\pi i}$ |
| 43. $e^{i\pi/3}$ | 44. $e^{-i\pi}$ |
| 45. $e^{2+i\pi}$ | 46. $e^{\pi+i}$ |

47. Use De Moivre's Theorem with $n = 3$ to express $\cos 3\theta$ and $\sin 3\theta$ in terms of $\cos \theta$ and $\sin \theta$.

48. Use Euler's formula to prove the following formulas for $\cos x$ and $\sin x$:

$$\cos x = \frac{e^{ix} + e^{-ix}}{2} \quad \sin x = \frac{e^{ix} - e^{-ix}}{2i}$$

49. If $u(x) = f(x) + ig(x)$ is a complex-valued function of a real variable x and the real and imaginary parts $f(x)$ and $g(x)$ are differentiable functions of x , then the derivative of u is defined to be $u'(x) = f'(x) + ig'(x)$. Use this together with Equation 5 to prove that if $F(x) = e^{rx}$, then $F'(x) = re^{rx}$ when $r = a + bi$ is a complex number.

50. (a) If u is a complex-valued function of a real variable, its indefinite integral $\int u(x) dx$ is an antiderivative of u . Evaluate

$$\int e^{(1+i)x} dx$$

(b) By considering the real and imaginary parts of the integral in part (a), evaluate the real integrals

$$\int e^x \cos x dx \quad \text{and} \quad \int e^x \sin x dx$$

(c) Compare with the method used in Example 5.5.4.

Glossary of Biological Terms

agar a gelatinous substance made of seaweed used in the culturing of microbes

Allee effect a phenomenon by which the per capita growth rate of a population is negative for small population sizes (for example, because of difficulty finding mates for reproduction)

allele a particular variant of the DNA that is found at a specific gene; the terms *gene* and *allele* are often used interchangeably but technically it is not correct to do so

allometry the relationship between the size of one part of an organism and the size of the organism itself; allometric growth refers to this relationship during the growth of an individual

amino acid the basic chemical subunit that makes up proteins; there are 20 different possible amino acids that are specified by codons during the process of protein synthesis

anesthesiology a branch of medicine dealing with the administration of drugs and anesthetics for controlling pain and sensation, as well as other life sustaining factors, particularly in the context of surgery

antibiotic concentration profile a graph depicting the concentration of an antibiotic in the body as a function of the time since it was administered

antigen any foreign substance or molecule that stimulates an immune response specific to it; often a surface molecule or cellular component of a pathogen

antigenic cartography the construction of a plot in which the binding strength of a virus (or other pathogen) to each of a set of n immunity molecules is depicted as a point in n -dimensional space

antigenic evolution evolution of the antigenic properties of a pathogen; it allows the pathogen to escape any previous immune response and thereby to reproduce until an immune response specific to the new form of its antigen develops

antigenic space an n -dimensional space whose coordinate axes represent the binding strength to each of n different immunity molecules

antigenicity the capacity of a substance or molecule to induce an immune response

antiserum serum containing immune system molecules against specific antigens

biomechanics the study of animal form in terms of mechanical structure and physical composition, especially as it relates to movement

CT/CAT scan computerized tomography or computerized axial tomography; an X-ray technique used for generating cross-sectional images of the internal parts of a body

carrying capacity the population size at which the per capita growth rate of the population is zero

chemotaxis a phenomenon by which an organism moves in the direction of a chemical gradient

codon a specific sequence of three consecutive nucleotides that codes for a particular amino acid during the process of protein synthesis

compensatory growth a phenomenon by which an individual exhibits an increase in growth rate after a period of food deprivation so as to “catch up” to the size that is typical for their age

cross-feeding a process by which one species or genotype feeds on the waste products of another

diploid a cell or organism that has two copies of each gene

DNA acronym for DeoxyriboNucleic Acid; a double-stranded nucleic acid molecule that contains the genetic information (that is, the genetic code) for many living organisms.

dose response relationship (curve) the relationship between the concentration or amount of a drug and its effect

dynamics a broad term referring to the change in a variable of interest over time; for example, “the dynamics of N ” refers to the way that N changes as a function of time

Einthoven’s triangle an imaginary equilateral triangle superimposed on a person’s chest, and used in vectorcardiography

electrocardiogram (ECG) a recording of the electrical activity of the heart over time

embolism a clot or blockage in a blood vessel

enzyme a type of molecule that catalyzes chemical reactions

fitness a measure of the survivorship and/or reproductive output of an organism

gait a pattern of animal locomotion such as walking, running, or galloping

gametocyte a specific kind of cell that is capable of producing either the male (sperm) or female (egg) cells required for sexual reproduction

gene the basic unit of heredity, typically identified as a specific location on the DNA; the terms *gene* and *allele* are often used interchangeably but technically it is not correct to do so

gene regulation a feedback process by which the expression of a gene is regulated by the products it generates

genome the entire set of genes of an organism

genome expression profile the measured levels of expression of a set of genes

genotype the collection of specific alleles within an organism

haploid a cell or organism that has one copy of each gene

Hardy-Weinberg law the statement that allele frequencies in a population remain constant over time in the absence of evolutionary processes such as natural selection or mutation

homeostatis the tendency of a biological system to maintain itself in a relatively stable state

inbreeding the interbreeding of genetically similar individuals

incidence in epidemiology, the rate at which individuals are getting a disease

infectious the state in which an infected individual is able to transmit a disease

island biogeography the study of the geographic distribution of organisms, particularly the species richness of isolated areas such as caves, islands, or lakes

kill curve a plot of the size of a bacterial population as a function of time after a dose of antibiotic is given

merozoite in malaria, a specific kind of cell formed during the stage of the life cycle when asexual cell division occurs

metamorphosis a large-scale change in the form and structure of an organism's body (and often its way of life) during development

metapopulation a group of spatially separated but interacting subpopulations

methylation (DNA) the attachment of chemical methyl groups to DNA

microarray a grid of microscopic gene fragments attached to a solid surface, and used to quantify the expression level of different genes

morphology a term referring to the form or structure of an organism

mRNA acronym for messenger RiboNucleic Acid; a type of RNA that serves as an intermediate molecule in the process of protein synthesis

mutate to change from one allele to another

mutation a term used to describe a new type of allele that arises when the "normal" allele mutates

neuron an electrically excitable cell that conducts electrical impulses (called nerve impulses) from one location in an organism to another

nucleotide a single letter of the genetic code that makes up the chemical structure of DNA and RNA

parasite an organism that obtains resources by living in or on another host organism (typically to the detriment of the host); the terms parasite and pathogen are sometimes used interchangeably

pathogen an organism that is capable of causing disease or illness in its host organism; the terms pathogen and parasite are sometimes used interchangeably

pathogenesis the development of a disease within an individual over time

per capita growth rate (per unit rate) the rate at which a population is growing divided by the size of the population at that time

pharmacokinetics the change in drug concentration within a patient during treatment

phenotype the observable properties of an organism in contrast to genotype, which is the collection of specific alleles within an organism and cannot be directly observed. An organism's genotype, in part, determines the organism's phenotype

photosynthesis the synthesis of organic carbon compounds from carbon dioxide, water, and sunlight

population genetics the study of how allele frequencies in a population change

prevalence in epidemiology, the proportion (or sometimes number) of individuals having a disease

renewable resource a biological resource that is self-reproducing, such as a forest or a fish population (as compared with nonrenewable resources such as oil)

reproduction number a term used in the study of infectious diseases; it is the number of new infections generated by a single infected individual when introduced into a population entirely susceptible to infection

RNA acronym for RiboNucleic Acid; a single-stranded nucleic acid molecule that plays an important role in translating the genetic code of DNA into proteins and also serves as the molecule of inheritance for some viruses

serum the clear portion of blood

sporozoite in malaria, a specific kind of cell involved in the stage of the life cycle when sexual cell division occurs; it is the cell type that is introduced into the human bloodstream by mosquitoes

substrate a substance upon which a chemical reaction occurs (often as a result of binding with an enzyme)

transcription an initial step in protein synthesis whereby the DNA sequence of a gene is transcribed into mRNA

translation a secondary step in protein synthesis whereby mRNA is read as a series of codons, each of which specifies an amino acid that goes into the synthesis of the protein

urea a chemical waste product produced by many organisms

vaccination a form of preventative medicine in which an individual is given an inactive form of a pathogen, called a vaccine, to stimulate an immune response that protects against subsequent infection

vaccine escape the antigenic evolution of a virus to the point where a vaccine no longer provides protection against it

vectorcardiography a method that records the direction and magnitude of electrical impulses generated by the heart

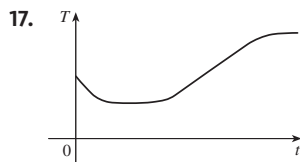
virulence a term referring to the severity, in terms of an individual's health, of an infectious disease

zygote the cell formed by the fusion of a male (sperm) and female (egg) cell during sexual reproduction

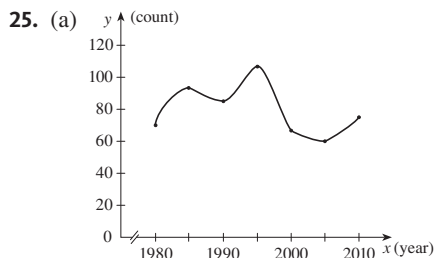
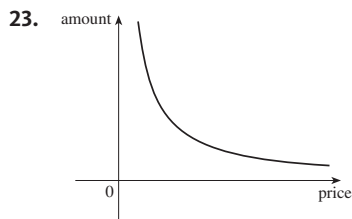
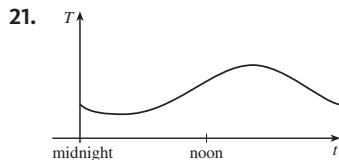
CHAPTER 1

EXERCISES 1.1 | page 13

1. Yes
 3. (a) 3 (b) -0.2 (c) $0, 3$ (d) -0.8
 (e) $[-2, 4]$, $[-1, 3]$ (f) $[-2, 1]$
 5. No 7. Yes, $[-3, 2]$, $[-3, -2] \cup [-1, 3]$
 9. (a) $\approx 13.8^\circ\text{C}$ (b) ≈ 1992 (c) $1910; 2006$
 (d) $[13.5, 14.5]$
 11. $[12:23 \text{ AM}, 12:52 \text{ AM}]$
 13. (a) $\approx 100, 134$ (b) 30°N or 30°S (c) Even function
 15. Diet, exercise, or illness

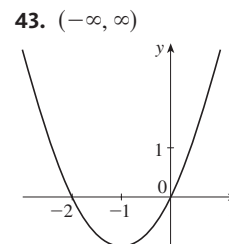
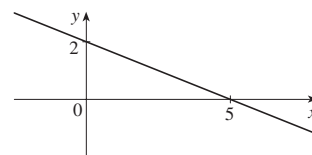


19. Nutrient consumption; growth; carrying capacity reached; death

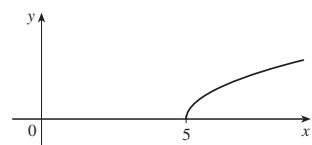


(b) $\approx 92,000$

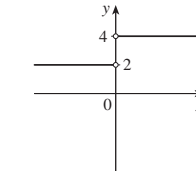
27. $12, 16, 3a^2 - a + 2, 3a^2 + a + 2, 3a^2 + 5a + 4, 6a^2 - 2a + 4, 12a^2 - 2a + 2, 3a^4 - a^2 + 2, 9a^4 - 6a^3 + 13a^2 - 4a + 4, 3a^2 + 6ah + 3h^2 - a - h + 2$
 29. $-3 - h$ 31. $-1/(ax)$
 33. $(-\infty, -3) \cup (-3, 3) \cup (3, \infty)$ 35. $(-\infty, \infty)$
 37. $(-\infty, 0) \cup (5, \infty)$ 39. $[0, 4]$
 41. $(-\infty, \infty)$



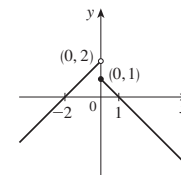
45. $[5, \infty)$



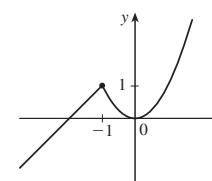
47. $(-\infty, 0) \cup (0, \infty)$



49. $(-\infty, \infty)$



51. $(-\infty, \infty)$

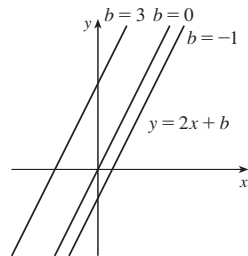


53. $A(L) = 10L - L^2, 0 < L < 10$
 55. $A(x) = \sqrt{3}x^2/4, x > 0$ 57. $S(x) = x^2 + (8/x), x > 0$
 59. $T(x) = \begin{cases} 75x & \text{if } 0 < x \leq 2 \\ 150 + 50(x - 2) & \text{if } x > 2 \end{cases}$
 61. ≈ 77 hours
 63. f is odd, g is even 65. (a) $(-5, 3)$ (b) $(-5, -3)$
 67. Odd 69. Neither 71. Even
 73. Even; odd; neither (unless $f = 0$ or $g = 0$)

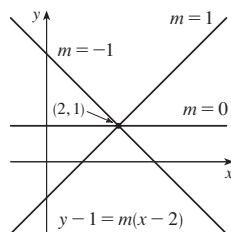
EXERCISES 1.2 | page 28

1. (a) Logarithmic (b) Root (c) Rational
 (d) Polynomial, degree 2 (e) Exponential (f) Trigonometric
 3. (a) h (b) f (c) g

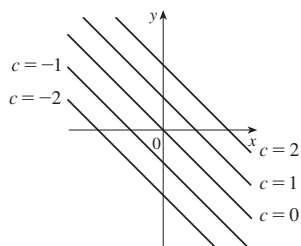
5. (a) $y = 2x + b$,
where b is the y -intercept.



- (b) $y = mx + 1 - 2m$,
where m is the slope.
(c) $y = 2x - 3$

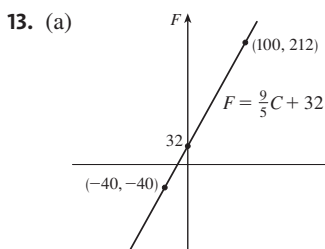


7. Their graphs have slope -1 .



9. $f(x) = -3x(x+1)(x-2)$

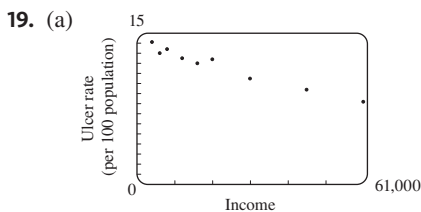
11. (a) 8.34, change in mg for every 1 year change
(b) 8.34 mg



- (b) $\frac{9}{5}$, change in $^{\circ}\text{F}$ for every 1°C change; 32, Fahrenheit temperature corresponding to 0°C

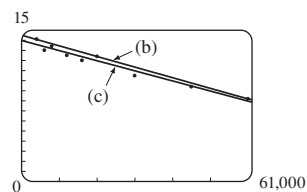
15. (a) $T = \frac{1}{6}N + \frac{307}{6}$ (b) $\frac{1}{6}$, change in $^{\circ}\text{F}$ for every chirp per minute change (c) 76°F

17. (a) Cosine (b) Linear



Linear model is appropriate.

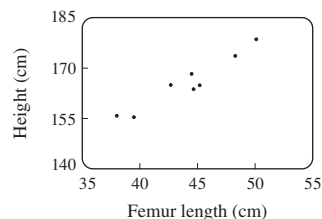
(b) $y = -0.000105x + 14.521$



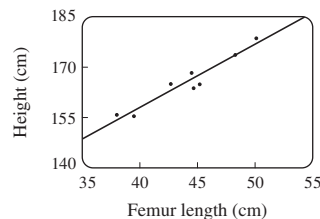
(c) $y = -0.00009979x + 13.951$ [See graph in part (b).]

- (d) About 11.5 per 100 population (e) About 6% (f) No

21. (a)



- (b) $H = 1.8807L + 82.6497$, where H is the height in centimeters and L is the femur length in centimeters

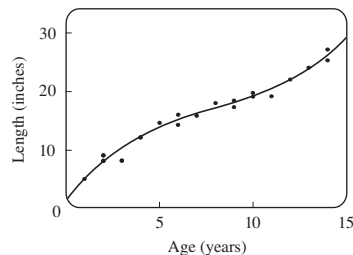


(c) ≈ 182.3 cm

23. Four times as bright

25. (a) $N = 3.1046A^{0.308}$ (b) 18

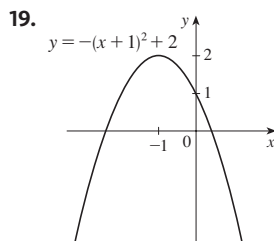
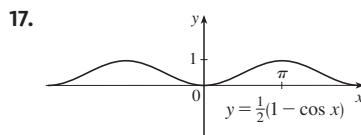
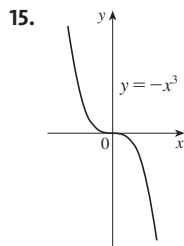
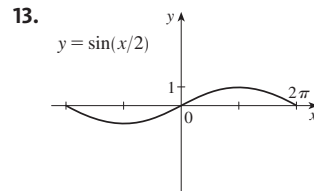
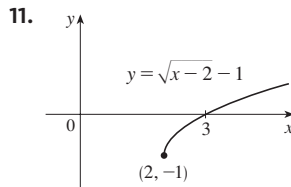
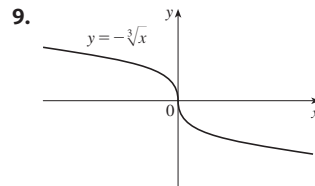
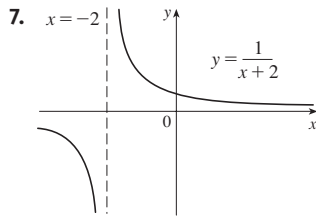
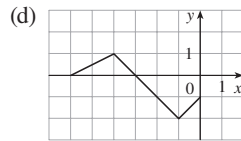
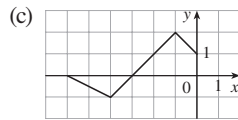
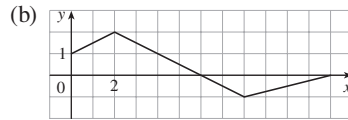
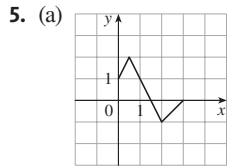
27. (a) $L = 0.0155A^3 - 0.3725A^2 + 3.9461A + 1.2108$



- (b) ≈ 13.6 in (c) ≈ 10.88 years

EXERCISES 1.3 | page 38

1. (a) $y = f(x) + 3$ (b) $y = f(x) - 3$ (c) $y = f(x - 3)$
(d) $y = f(x + 3)$ (e) $y = -f(x)$ (f) $y = f(-x)$
(g) $y = 3f(x)$ (h) $y = \frac{1}{3}f(x)$
3. (a) 3 (b) 1 (c) 4 (d) 5 (e) 2



21. $L(t) = 12 + 2 \sin \left[\frac{2\pi}{365} (t - 80) \right]$

23. $D(t) = 5 \cos \left(\frac{\pi}{6} (t - 6.75) \right) + 7$

25. $f(t) = 30 \sin \left(\frac{2\pi}{3} t \right) + 50$

27. (a) $(f + g)(x) = x^3 + 5x^2 - 1, (-\infty, \infty)$

(b) $(f - g)(x) = x^3 - x^2 + 1, (-\infty, \infty)$

(c) $(fg)(x) = 3x^5 + 6x^4 - x^3 - 2x^2, (-\infty, \infty)$

(d) $(f/g)(x) = \frac{x^3 + 2x^2}{3x^2 - 1}, \{x \mid x \neq \pm 1/\sqrt{3}\}$

29. (a) $(f \circ g)(x) = 4x^2 + 4x, (-\infty, \infty)$

(b) $(g \circ f)(x) = 2x^2 - 1, (-\infty, \infty)$

(c) $(f \circ f)(x) = x^4 - 2x^2, (-\infty, \infty)$

(d) $(g \circ g)(x) = 4x + 3, (-\infty, \infty)$

31. (a) $(f \circ g)(x) = 1 - 3 \cos x, (-\infty, \infty)$

(b) $(g \circ f)(x) = \cos(1 - 3x), (-\infty, \infty)$

(c) $(f \circ f)(x) = 9x - 2, (-\infty, \infty)$

(d) $(g \circ g)(x) = \cos(\cos x), (-\infty, \infty)$

33. (a) $(f \circ g)(x) = \frac{2x^2 + 6x + 5}{(x + 2)(x + 1)}, \{x \mid x \neq -2, -1\}$

(b) $(g \circ f)(x) = \frac{x^2 + x + 1}{(x + 1)^2}, \{x \mid x \neq -1, 0\}$

(c) $(f \circ f)(x) = \frac{x^4 + 3x^2 + 1}{x(x^2 + 1)}, \{x \mid x \neq 0\}$

(d) $(g \circ g)(x) = \frac{2x + 3}{3x + 5}, \{x \mid x \neq -2, -5/3\}$

35. $(f \circ g \circ h)(x) = 3 \sin(x^2) - 2$

37. $(f \circ g \circ h)(x) = \sqrt{x^6 + 4x^3 + 1}$

39. $g(x) = 2x + x^2, f(x) = x^4$

41. $g(x) = \sqrt[3]{x}, f(x) = x/(1 + x)$

43. $g(t) = t^2, f(t) = \sec t \tan t$

45. $h(x) = \sqrt{x}, g(x) = x - 1, f(x) = \sqrt{x}$

47. $h(x) = \sqrt{x}, g(x) = \sec x, f(x) = x^4$

49. (a) 4 (b) 3 (c) 0 (d) Does not exist; $f(6) = 6$ is not in the domain of g . (e) 4 (f) -2

51. (a) $r(t) = 60t$ (b) $(A \circ r)(t) = 3600\pi t^2$; the area of the circle as a function of time

53. (a) $s = \sqrt{d^2 + 36}$ (b) $d = 30t$

(c) $(f \circ g)(t) = \sqrt{900t^2 + 36}$; the distance between the lighthouse and the ship as a function of the time elapsed since noon

55. (a) $d(t) = gt$ (in mm)

(b) $(P \circ S \circ d)(t) = kg^2\pi t^2$; the rate of enzyme production as a function of time

57. Yes; $m_1 m_2$

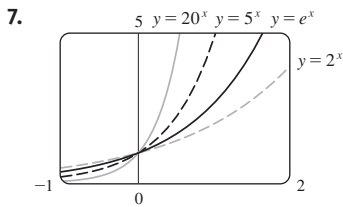
59. No; h is odd; h is even

EXERCISES 1.4 | page 50

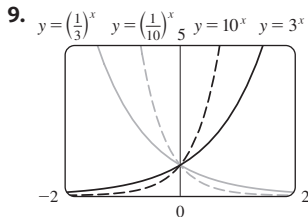
1. (a) 4 (b) $x^{-4/3}$ 3. (a) $16b^{12}$ (b) $648y^7$

5. (a) $f(x) = b^x, b > 0$ (b) $\mathbb{R}$ (c) $(0, \infty)$

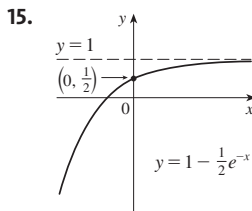
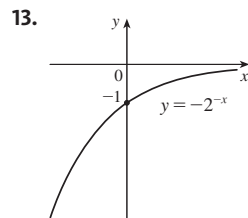
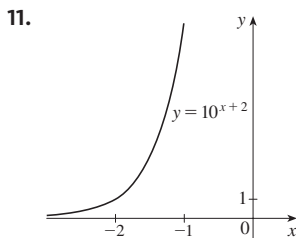
(d) See Figures 5(c), 5(b), and 5(a), respectively.



All approach 0 as $x \rightarrow -\infty$, all pass through $(0, 1)$, and all are increasing. The larger the base, the faster the rate of increase.



The functions with base greater than 1 are increasing and those with base less than 1 are decreasing. The latter are reflections of the former about the y -axis.

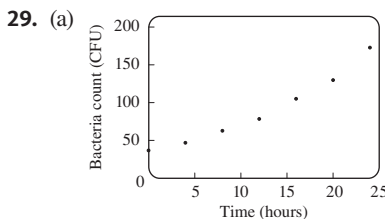


17. (a) $y = e^x - 2$ (b) $y = e^{x-2}$ (c) $y = -e^x$

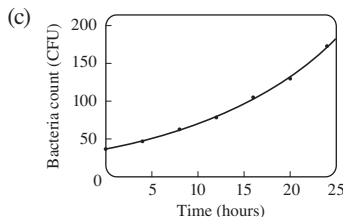
(d) $y = e^{-x}$ (e) $y = -e^{-x}$

19. (a) $(-\infty, -1) \cup (-1, 1) \cup (1, \infty)$ (b) $(-\infty, \infty)$

21. $f(x) = 3 \cdot 2^x$ 27. At $x \approx 35.8$



(b) $f(t) = (36.78) \cdot (1.07)^t$



≈ 10.8 hours for bacteria count to double

31. (a) 25 mg (b) $200 \cdot 2^{-t/5}$ (c) 10.9 mg (d) 38.2 days

33. ≈ 3.5 days

35. $P = 2614.086(1.01693)^t$, where $t = 0$ in 1950; 1993: $P \approx 5381$ million; 2020: $P \approx 8466$ million

EXERCISES 1.5 | page 65

1. (a) See Definition 1.

(b) It must pass the Horizontal Line Test.

3. No 5. No 7. Yes 9. No

11. Yes 13. No

15. (a) 6 (b) 3 17. 0

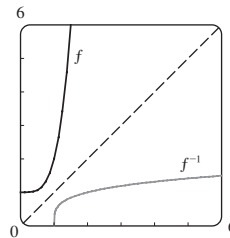
19. $F = \frac{9}{5}C + 32$; the Fahrenheit temperature as a function of the Celsius temperature; $[-273.15, \infty)$

21. $y = \frac{1}{3}(x-1)^2 - \frac{2}{3}, x \geq 1$

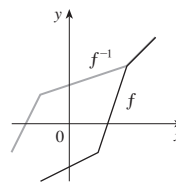
23. $y = \frac{1}{2}(1 + \ln x)$

25. $y = e^x - 3$

27. $f^{-1}(x) = \sqrt[4]{x-1}$



29.



31. (a) $f^{-1}(x) = \sqrt{1-x^2}, 0 \leq x \leq 1$; f^{-1} and f are the same function. (b) Quarter-circle in the first quadrant

33. (a) It's defined as the inverse of the exponential function with base b , that is, $\log_b x = y \iff b^y = x$.

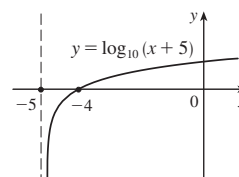
(b) $(0, \infty)$ (c) $\mathbb{R}$ (d) See Figure 11.

35. (a) 3 (b) -3

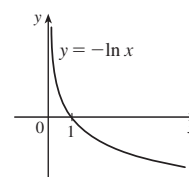
37. (a) 3 (b) -2 39. $\ln 1215$

41. $\ln \frac{\sqrt{x}}{x+1}$ 43. About 1,084,588 mi

45. (a)



(b)



47. (a) $\frac{1}{4}(7 - \ln 6)$ (b) $\frac{1}{3}(e^2 + 10)$

49. (a) $5 + \log_2 3$ or $5 + (\ln 3)/\ln 2$ (b) $\frac{1}{2}(1 + \sqrt{1 + 4e})$

51. (a) $0 < x < 1$ (b) $x > \ln 5$

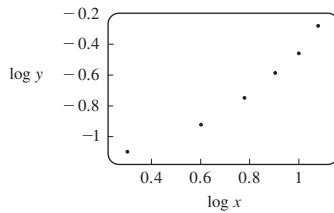
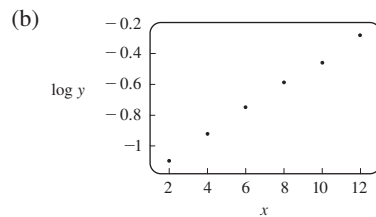
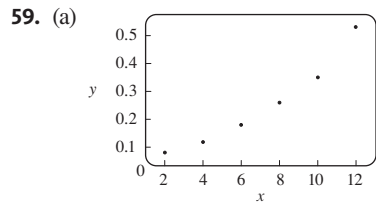
53. (a) $-\frac{32941}{340} \ln\left(\frac{0.60}{1.65}\right) \approx 98.0$ minutes

(b) $T = -\frac{V}{K} \ln\left(\frac{c(T)}{c_0}\right)$

55. (a) $(\ln 3, \infty)$ (b) $f^{-1}(x) = \ln(e^x + 3); \mathbb{R}$

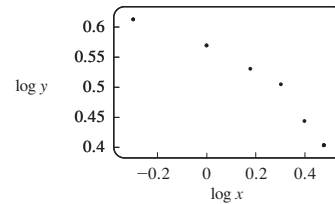
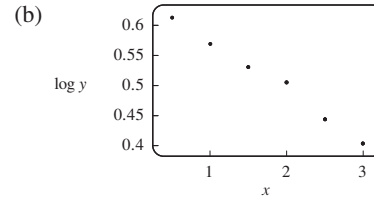
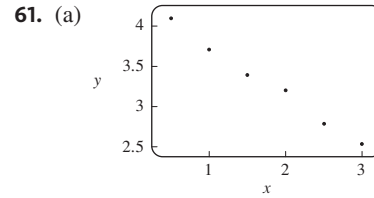
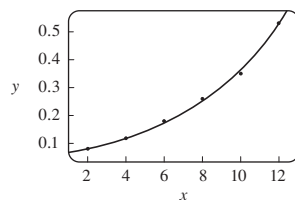
57. (a) $t = \frac{\log(n/500)}{\log 4}$; the inverse function gives the number of hours that have passed when the population size reaches n

(b) $\frac{\log 20}{\log 4} \approx 2.16$ h



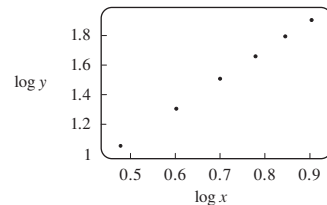
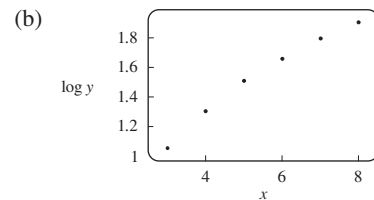
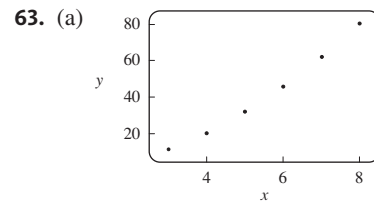
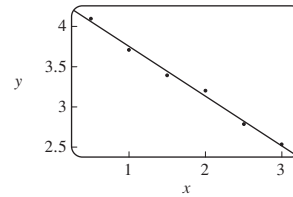
(c) Exponential

(d) $y = 0.056769(1.204651)^x$



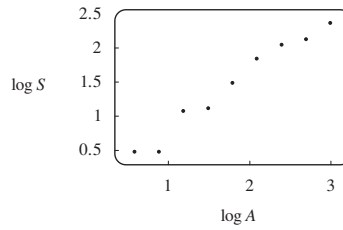
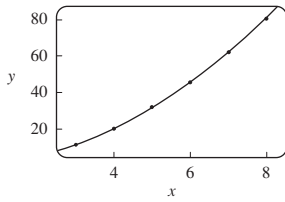
(c) Linear

(d) $y = -0.618857x + 4.368000$



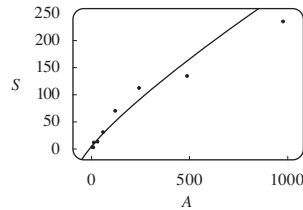
(c) Power model is appropriate.

(d) $y = 1.260294x^{2.002959}$

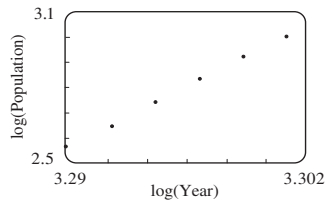
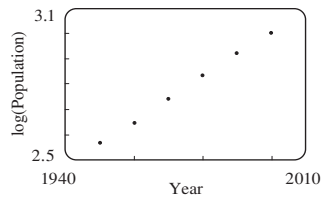
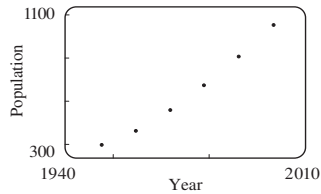


Power model is appropriate.

(b) $S = 0.881518A^{0.841701}$



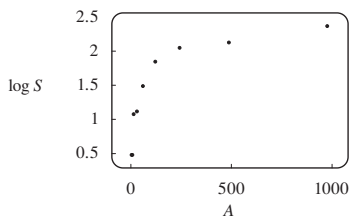
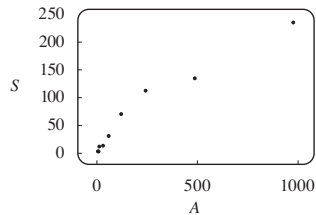
65. (a)



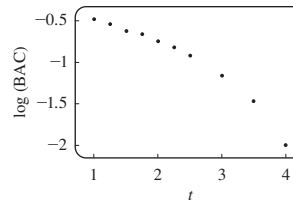
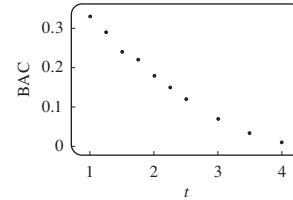
Exponential model is appropriate.

(b) $P = (2.276131 \cdot 10^{-15}) \cdot (1.020529)^Y$, where P is the population in millions and Y is the year.(c) $P \approx 1247$ million. Model overestimates actual population by 74 million. It does not generalize well to future population growth.

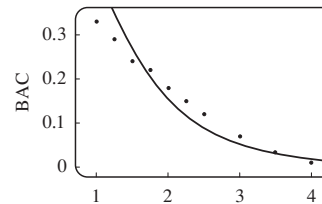
67. (a)



69. (a)



(b) $BAC = 1.343328(0.338676)^t$

The exponential function overestimates BAC for small values of t .

(c) After 2.6 h

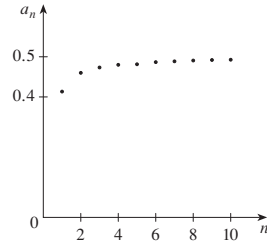
■ EXERCISES 1.6 | page 76

1. $\left\{1, \frac{4}{5}, \frac{3}{5}, \frac{8}{17}, \frac{5}{13}, \dots\right\}$

3. $\left\{\frac{1}{5}, -\frac{1}{25}, \frac{1}{125}, -\frac{1}{625}, \frac{1}{3125}, \dots\right\}$

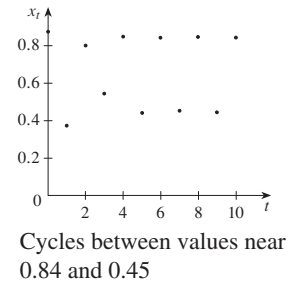
5.

n	$a_n = \frac{3n}{1+6n}$
1	0.4286
2	0.4615
3	0.4737
4	0.4800
5	0.4839
6	0.4865
7	0.4884
8	0.4898
9	0.4909
10	0.4918



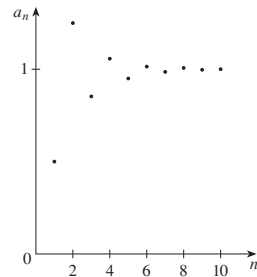
29.

t	x_t
0	0.8750
1	0.3741
2	0.8008
3	0.5456
4	0.8479
5	0.4411
6	0.8431
7	0.4523
8	0.8472
9	0.4427
10	0.8438



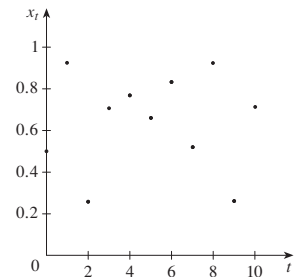
7.

n	$a_n = 1 + (-\frac{1}{2})^n$
1	0.5000
2	1.2500
3	0.8750
4	1.0625
5	0.9688
6	1.0156
7	0.9922
8	1.0039
9	0.9980
10	1.0010



31.

t	x_t
0	0.5000
1	0.9250
2	0.2567
3	0.7060
4	0.7681
5	0.6591
6	0.8313
7	0.5189
8	0.9237
9	0.2608
10	0.7134



Cycles irregularly among a range of values between 0.26 and 0.92

9. $a_n = 1/(2n-1)$ 11. $a_n = -3(-\frac{2}{3})^{n-1}$

13. $a_n = (-1)^{n+1} \frac{n^2}{n+1}$ 15. $\{1, 2, 7, 32, 157, 782, \dots\}$

17. $\left\{2, \frac{2}{3}, \frac{2}{5}, \frac{2}{7}, \frac{2}{9}, \frac{2}{11}, \dots\right\}$

19. $\{1, 3^{1/2}, 3^{3/4}, 3^{7/8}, 3^{15/16}, 3^{31/32}, \dots\}$

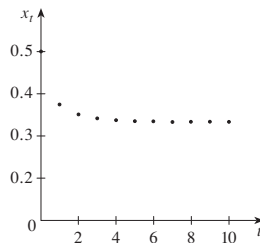
21. $\{2, 1, -1, -2, -1, 1, \dots\}$

25. (a) As t increases, N_t approaches 0. (b) $N_t = 1$ for all t .

(c) The sequence grows indefinitely as t increases.

27.

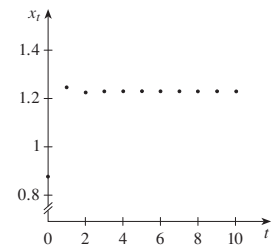
t	x_t
0	0.5000
1	0.3750
2	0.3516
3	0.3419
4	0.3375
5	0.3354
6	0.3344
7	0.3338
8	0.3336
9	0.3335
10	0.3334



Approaches 1/3

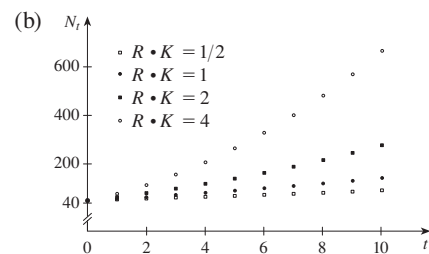
33.

t	x_t
0	0.8750
1	1.2475
2	1.2254
3	1.2306
4	1.2294
5	1.2297
6	1.2296
7	1.2296
8	1.2296
9	1.2296
10	1.2296



Ricker: approaches 1.2296;
Logistic: cycles between values

35. (a) $N_{t+1} = N_t + RK\sqrt{N_t}$; R and K constants



37. (a) $n_{t+1} = b(1-d)n_t + m$

(b) $n_{t+1} = (1-d)(bn_t + m)$

(c) Part (a) recursion formula

39. $p_{t+1} = \frac{\alpha p_t}{\alpha p_t + 1 - p_t}$

■ CHAPTER 1 REVIEW | page 81

True-False Quiz

1. False 3. False 5. True 7. False 9. True 11. False

Exercises

1. (a) 2.7 (b) 2.3, 5.6 (c) $[-6, 6]$ (d) $[-4, 4]$ (e) $[-4, 4]$ (f) No; it fails the Horizontal Line Test.

(g) Odd; its graph is symmetric about the origin.

3. (a) ≈ -36 m

(b) 18,000 years ago; 121,000 years ago

(c) $[-114, 8]$

(d) Sea level drops correspond to periods of glaciation.

5. $2a + h - 2$ 7. $(-\infty, \frac{1}{3}) \cup (\frac{1}{3}, \infty)$, $(-\infty, 0) \cup (0, \infty)$ 9. $(-6, \infty)$, $\mathbb{R}$

11. (a) Shift the graph 8 units upward.

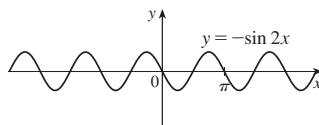
(b) Shift the graph 8 units to the left.

(c) Stretch the graph vertically by a factor of 2, then shift it 1 unit upward.

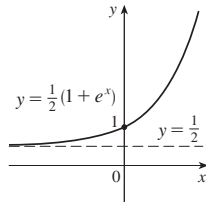
(d) Shift the graph 2 units to the right and 2 units downward.

(e) Reflect the graph about the x -axis.(f) Reflect the graph about the line $y = x$ (assuming f is one-to-one).

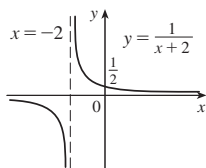
13.



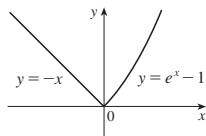
15.



17.



19.



21. (a) $(f \circ g)(x) = \ln(x^2 - 9)$, $(-\infty, -3) \cup (3, \infty)$

(b) $(g \circ f)(x) = (\ln x)^2 - 9$, $(0, \infty)$

(c) $(f \circ f)(x) = \ln \ln x$, $(1, \infty)$

(d) $(g \circ g)(x) = (x^2 - 9)^2 - 9$, $(-\infty, \infty)$

23. $y = 0.2493x - 423.4818$; about 77.6 years

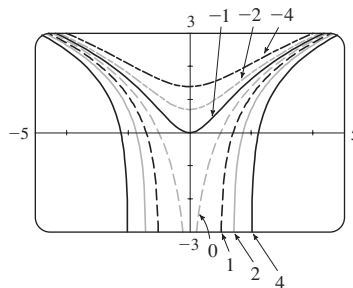
25. 1 27. (a) 9 (b) 2

29. (a) $\frac{1}{16}$ g (b) $m(t) = 2^{-t/4}$

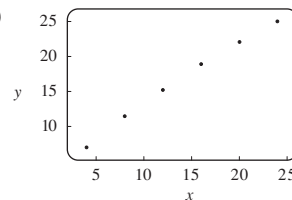
(c) $t(m) = -4 \log_2 m$; the time elapsed when there are m grams of ^{100}Pd

(d) About 26.6 days

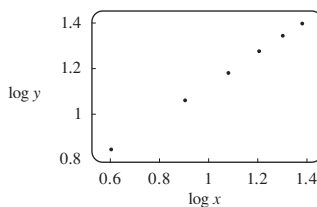
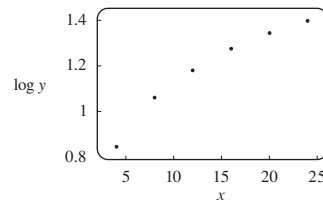
31.

For $c < 0$, f is defined everywhere. As c increases, the dip at $x = 0$ becomes deeper. For $c \geq 0$, the graph has asymptotes at $x = \pm\sqrt{c}$.

33. (a)

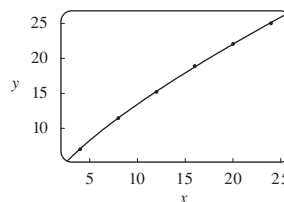


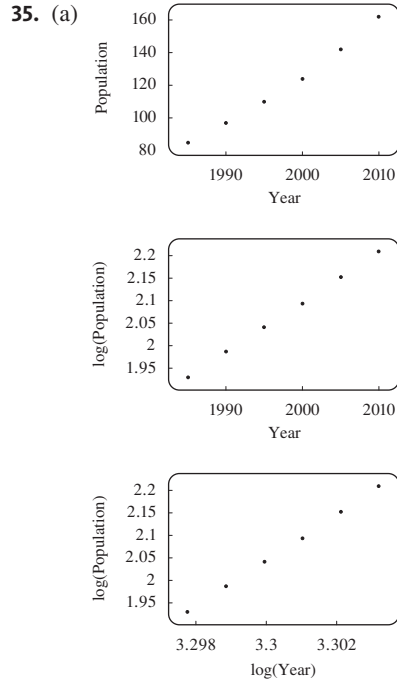
(b)



(c) Power

(d) $y = 2.608377x^{0.712277}$



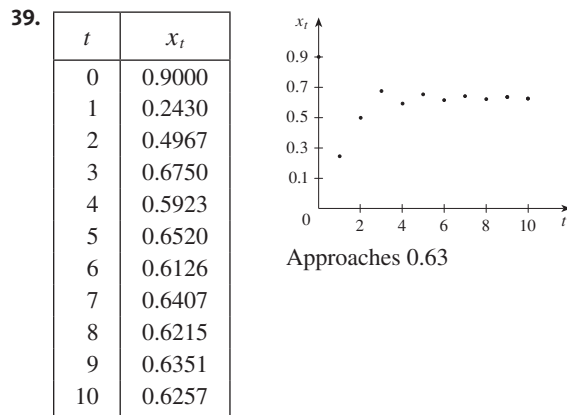


Exponential or power model

(b) $P = (6.6326 \cdot 10^{-21}) \cdot (1.025977)^Y$, where P is the population in millions and Y is the year.

(c) 153 million; 209 million

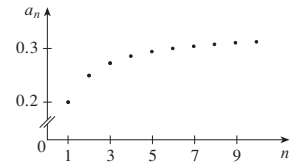
37. $\{3, 6, 13, 28, 59, 122, \dots\}$



3. The sequence likely has a nonzero limit as $t \rightarrow \infty$ because human physiology will ultimately limit how fast a human can sprint 100 meters. This means that there is a certain world record time that athletes will never surpass.

5.

n	a_n
1	0.2000
2	0.2500
3	0.2727
4	0.2857
5	0.2941
6	0.3000
7	0.3043
8	0.3077
9	0.3103
10	0.3125

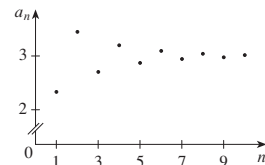


The sequence appears to converge to a number between 0.3 and 0.35.

$\lim_{n \rightarrow \infty} a_n = \frac{1}{3}$. This agrees with value predicted from the data.

7.

n	a_n
1	2.3333
2	3.4444
3	2.7037
4	3.1975
5	2.8683
6	3.0878
7	2.9415
8	3.0390
9	2.9740
10	3.0173



The sequence appears to converge to a number near 3.

$\lim_{n \rightarrow \infty} a_n = 3$. This agrees with value predicted from the data.

9. 0 11. 2 13. $\frac{5}{7}$

15. 1 17. Diverges

19. Diverges 21. Diverges

23. $\ln 2$ 25. 0

27. $\lim_{n \rightarrow \infty} a_n = 2$

n	a_n
1	1.0000
2	1.5000
3	1.7500
4	1.8750
5	1.9375
6	1.9688
7	1.9844
8	1.9922

CHAPTER 2

EXERCISES 2.1 | page 99

1. (a) A sequence is an ordered list of numbers. It can also be defined as a function whose domain is the set of positive integers.
- (b) The terms a_n approach 8 as n becomes large.
- (c) The terms a_n become large as n becomes large.

29. Divergent

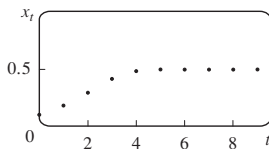
n	a_n
1	2.0000
2	3.0000
3	5.0000
4	9.0000
5	17.0000
6	33.0000
7	65.0000
8	129.0000

31. $\lim_{n \rightarrow \infty} a_n = 2$

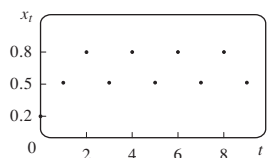
n	a_n
1	1.0000
2	3.0000
3	1.5000
4	2.4000
5	1.7647
6	2.1702
7	1.8926
8	2.0742

33. $\lim_{n \rightarrow \infty} a_n = 2$

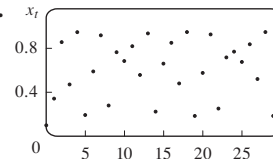
n	a_n
1	1.0000
2	1.7321
3	1.9319
4	1.9829
5	1.9957
6	1.9989
7	1.9997
8	1.9999

35. (a) 120 mg, 124 mg (b) $Q_{n+1} = 100 + (0.20)Q_n$
(c) $Q_n = 125(1 - 0.20^n)$ (d) $\lim_{n \rightarrow \infty} Q_n = 125$ mg37. (a) 157.875 mg; $\frac{3000}{19}(1 - 0.05^n)$ (b) 157.895 mg39. (a) $x = 1$ (b) 1 (c) 2
(d) All rational numbers with a terminating decimal representation, except 0.41. $\frac{8}{9}$ 43. $\frac{838}{333}$ 45. 5063/330047. Converges to $1/2$.
 $\lim_{t \rightarrow \infty} x_t = 1/2$.

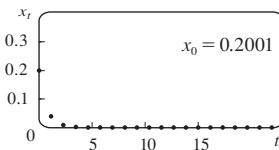
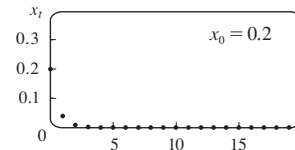
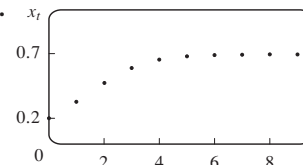
49. Diverges



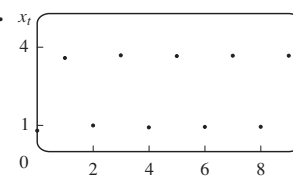
51. Diverges



53.

The plots show that the solutions are nearly identical, converging to zero as t increases.55. Converges to approximately 0.7; $\lim_{t \rightarrow \infty} x_t = \ln 2$ 

57. Diverges

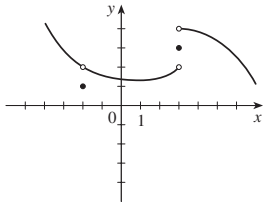
59. The removed area of the Sierpinski carpet after the n th step of construction is $A_n = 1 - (\frac{8}{9})^n$ so, $\lim_{n \rightarrow \infty} A_n = 1$ implying that the Sierpinski carpet has zero area.**EXERCISES 2.2 | page 109**1. (a) As x becomes large, $f(x)$ approaches 5.(b) As x becomes large negative, $f(x)$ approaches 3.3. 0 5. 0 7. $\frac{3}{2}$ 9. $-\frac{1}{2}$ 11. ∞ 13. -1 15. 4 17. $\frac{1}{6}$ 19. 2 21. ∞ 23. $-\infty$ 25. 127. $-\frac{1}{2}$ 29. c is the concentration at which the growth rate is half that of the maximum possible value. This is often referred to as the half-saturation coefficient.

31. 0. As the mortality rate increases, the number of new infections approaches zero.

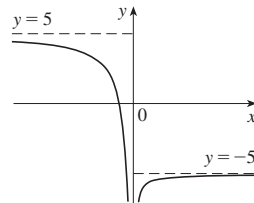
33. 8×10^7 . In the long run the biomass of the Pacific halibut will tend to 8×10^7 kg.35. $x > 9.21$ 37. (a) v^* (b) 3.52 s**EXERCISES 2.3 | page 122**1. As x approaches 2, the value of $f(x)$ approaches 5; yes3. (a) $\lim_{x \rightarrow -3} f(x) = \infty$ means that the values of $f(x)$ can be made arbitrarily large (as large as we please) by taking x sufficiently close to -3 (but not equal to -3).(b) $\lim_{x \rightarrow 4} f(x) = -\infty$ means that the values of $f(x)$ can be made arbitrarily large negative by taking x sufficiently close to 4 through values larger than 4.

5. (a) 2 (b) 1 (c) 4 (d) Does not exist (e) 3
 7. (a) 260 (b) 254 (c) Does not exist (d) 254
 (e) 258 (f) Does not exist (g) 258
 (h) On June 3 ($t = 2$), the population decreased by 6. This could have been a result of deaths, emigration, or a combination of the two. On June 5 ($t = 4$), the population increased by 4. This could have been a result of births, immigration, or a combination of the two.
 9. (a) $-\infty$ (b) $-\infty$ (c) ∞ (d) 2 (e) -1
 (f) Vertical: $x = 0, x = 2$; horizontal: $y = -1, y = 2$

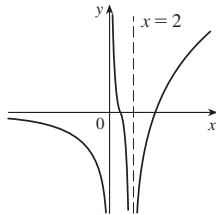
11.



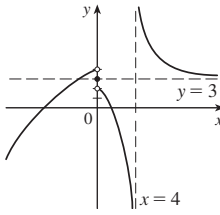
13.



15.



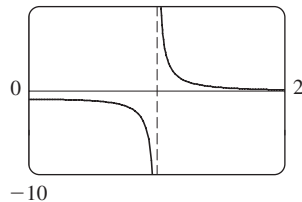
17.



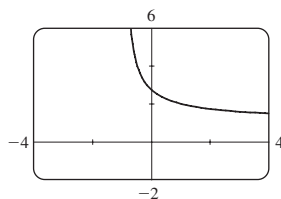
19. $\frac{2}{3}$ 21. 5 23. $\frac{1}{4}$ 25. $\frac{3}{5}$
 27. (a) -1.5 (b)

x	$f(x)$
± 0.1	-1.493759
± 0.01	-1.499938
± 0.001	-1.499999
± 0.0001	-1.500000

29. $-\infty$ 31. ∞ 33. $-\infty$ 35. $-\infty$ 37. ∞
 39. (a) $\lim_{x \rightarrow 1^-} f(x) = -\infty$ and $\lim_{x \rightarrow 1^+} f(x) = \infty$
 (c) 10



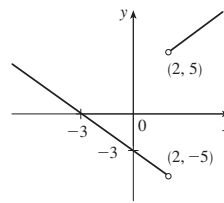
41. (a) 2.71828 (b)



43. No matter how many times we zoom in toward the origin, the graph appears to consist of almost-vertical lines. This indicates more and more frequent oscillations as $x \rightarrow 0$.

EXERCISES 2.4 | page 135

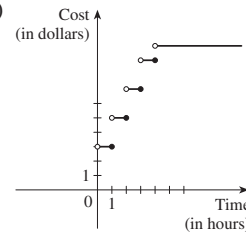
1. (a) -6 (b) -8 (c) 2 (d) -6
 (e) Does not exist (f) 0
 3. 59 5. $\frac{3}{2}$ 7. $\pi/2$
 9. 4 11. Does not exist
 13. $\frac{6}{5}$ 15. 8 17. $\frac{1}{12}$ 19. $-\frac{1}{16}$
 21. $\frac{1}{128}$ 23. $-\frac{1}{2}$ 25. (a), (b) $\frac{2}{3}$
 29. 7 33. 6 35. Does not exist
 37. (a) (i) 5 (ii) -5 (b) Does not exist
 (c)



39. 3 41. 3 43. $\frac{1}{2}$ 49. 8 51. 15; -1

EXERCISES 2.5 | page 147

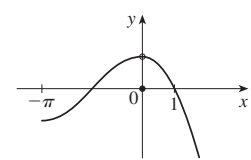
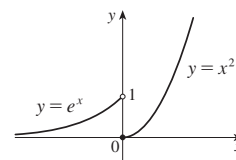
1. $\lim_{x \rightarrow 4} f(x) = f(4)$
 3. (a) $f(-4)$ is not defined and $\lim_{x \rightarrow a} f(x)$ does not exist [for $a = -2, 2$, and 4]
 (b) -4 , neither; -2 , left; 2 , right; 4 , right
 5. 7.
 9. (a) Discontinuous at $t = 12, 24, 36$ (b) Jump discontinuities
 11. (a)



- (b) Discontinuous at $t = 1, 2, 3, 4$
 13. 6

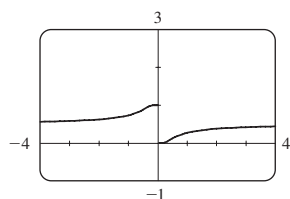
17. $\lim_{x \rightarrow 0} f(x)$ does not exist.

19. $\lim_{x \rightarrow 0} f(x) \neq f(0)$



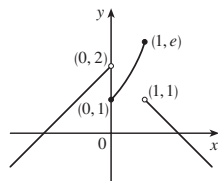
21. $[\frac{1}{2}, \infty)$ 23. $(-\infty, \infty)$
 25. $(-\infty, -1) \cup (1, \infty)$

27. $x = 0$



29. $\frac{7}{3}$ 31. 1

35. 0, right; 1, left



37. $\frac{2}{3}$ 45 (b) (0.86, 0.87) 47. (b) 70.347 49. Yes

CHAPTER 2 REVIEW | page 150

True-False Quiz

1. True 3. False 5. True 7. False 9. True

11. True 13. False 15. True

Exercises

1. $\frac{1}{2}$ 3. Diverges 5. $\frac{9}{2}$ 7. $\frac{4111}{3330}$

9. (a) (i) 3 (ii) 0 (iii) Does not exist (iv) 2

(v) ∞ (vi) $-\infty$ (vii) 4 (viii) -1

(b) $y = 4$, $y = -1$ (c) $x = 0$, $x = 2$ (d) -3, 0, 2, 4

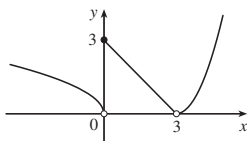
11. $-\frac{1}{5}$ 13. 1 15. $\frac{3}{2}$ 17. 3 19. ∞ 21. $\frac{4}{7}$

23. $-\infty$ 25. $\frac{1}{2}$ 27. 2

29. 0.50. As the concentration grows larger, the enzymatic reaction rate will approach 0.50.

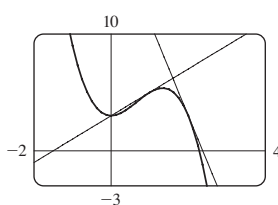
31. (a) (i) 3 (ii) 0 (iii) Does not exist (iv) 0 (v) 0 (vi) 0

(b) At 0 and 3 (c)



9. (a) $8a - 6a^2$ (b) $y = 2x + 3$, $y = -8x + 19$

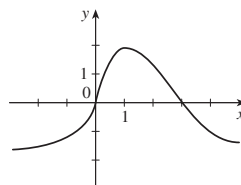
(c)



11. $g'(0)$, 0, $g'(4)$, $g'(2)$, $g'(-2)$

13. $f(2) = 3$; $f'(2) = 4$

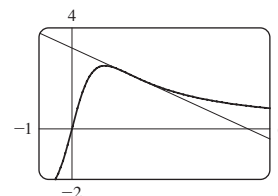
15.



17. $y = 3x - 1$

19. (a) $-\frac{3}{5}$; $y = -\frac{3}{5}x + \frac{16}{5}$

(b)



21. $6a - 4$ 23. $\frac{5}{(a + 3)^2}$

25. $-\frac{1}{\sqrt{1 - 2a}}$ 27. -24 ft/s

29. $-2/a^3$ m/s; -2 m/s; $-\frac{1}{4}$ m/s; $-\frac{2}{27}$ m/s

31. (a) (i) 3.26 million/year (ii) 3.18 million/year

(iii) 2.72 million/year (iv) 2.61 million/year

(b) $P'(2000) \approx 2.95$ million/year; the US population was growing at a rate of 2.95 million people per year in 2000.

33. (a) (i) -0.15 (mg/mL)/h (ii) -0.12 (mg/mL)/h

(iii) -0.12 (mg/mL)/h (iv) -0.11 (mg/mL)/h

(b) $C'(2.0) \approx -0.12$ (mg/mL)/h, meaning that the blood alcohol concentration was decreasing at a rate of 0.12 mg/mL per hour after 2.0 hours.

35. $T'(12) \approx 2.75^\circ\text{F/h}$; $T'(12)$ is the rate of change of temperature 12 hours after midnight on May 7, 2012.

37. Greater (in magnitude)

39. (a) The rate at which the cost is changing per ounce of gold produced; dollars per ounce

(b) When the 800th ounce of gold is produced, the cost of production is \$17/oz.

(c) Decrease in the short term; increase in the long term

41. (a) The rate at which the oxygen solubility changes with respect to the water temperature; (mg/L)/ $^\circ\text{C}$

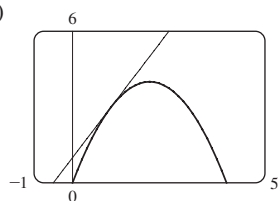
(b) $S'(16) \approx -0.25$; as the temperature increases past 16°C , the oxygen solubility is decreasing at a rate of 0.25 (mg/L)/ $^\circ\text{C}$.

CHAPTER 3

EXERCISES 3.1 | page 165

1. (a) $\frac{f(x) - f(3)}{x - 3}$ (b) $\lim_{x \rightarrow 3} \frac{f(x) - f(3)}{x - 3}$

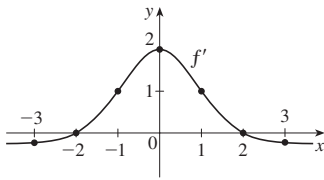
3. (a) 2 (b) $y = 2x + 1$ (c)



5. $y = -8x + 12$ 7. $y = \frac{1}{2}x + \frac{1}{2}$

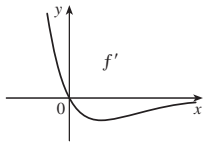
EXERCISES 3.2 | page 178

1. (a) -0.2 (b) 0 (c) 1 (d) 2
 (e) 1 (f) 0 (g) -0.2

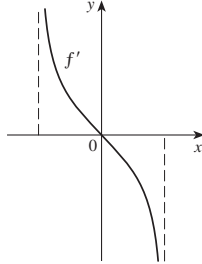


3. (a) II (b) IV (c) I (d) III

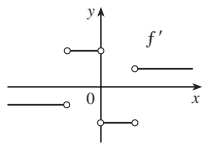
5.



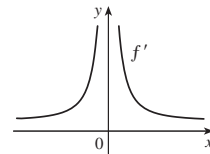
7.



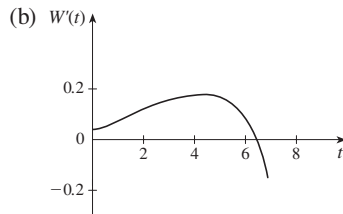
9.



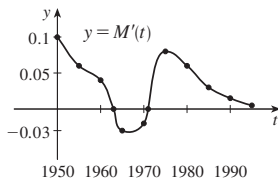
11.



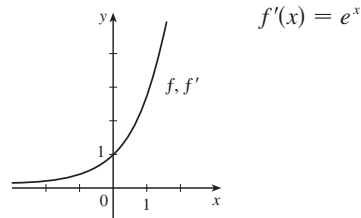
13. (a) $W'(t)$ is the rate of change of average body weight with respect to time for tadpoles raised in a density of 80 tadpoles/L.



15. 1963 to 1971



17.

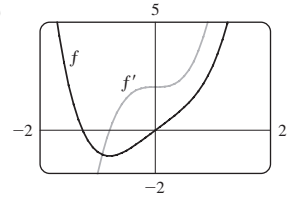


19. (a) 0, 1, 2, 4 (b) -1, -2, -4 (c) $f'(x) = 2x$
 21. $f'(x) = \frac{1}{2}$, $\mathbb{R}$, $\mathbb{R}$ 23. $f'(t) = 5 - 18t$, $\mathbb{R}$, $\mathbb{R}$
 25. $f'(x) = 2x - 6x^2$, $\mathbb{R}$, $\mathbb{R}$
 27. $g'(x) = 1/\sqrt{1 + 2x}$, $[-\frac{1}{2}, \infty)$, $(-\frac{1}{2}, \infty)$

29. $G'(t) = \frac{4}{(t+1)^2}$, $(-\infty, -1) \cup (-1, \infty)$, $(-\infty, -1) \cup (-1, \infty)$

31. $f'(x) = 4x^3$, $\mathbb{R}$, $\mathbb{R}$

33. (a) $f'(x) = 4x^3 + 2$ (b)



35. (a) $N'(t)$ is the rate of change of the number of malarial parasites with respect to time. Its units are (number of parasites/ μL)/day.

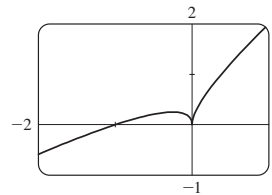
(b)

t	$N'(t)$
2	6,261
3	12,152
4	179,791
5	1,095,390
6	3,188,035

37. -4 (corner); 0 (discontinuity)

39. -1 (vertical tangent); 4 (corner)

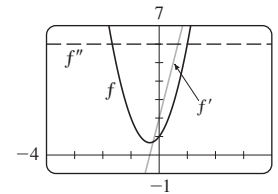
41.



Differentiable at -1;
not differentiable at 0

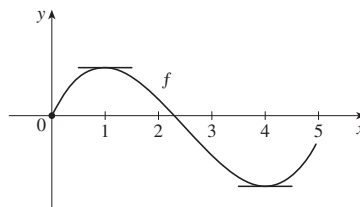
43. $a = f$, $b = f'$, $c = f''$

45. $6x + 2$; 6



47. (a) Increasing on (0, 1), (4, 5); decreasing on (1, 4)

(b)


EXERCISES 3.3 | page 192

1. (a) See Definition of the number e (page 189).

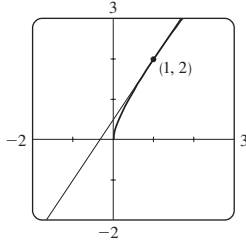
- (b) 0.99, 1.03; $2.7 < e < 2.8$

3. $f'(x) = 0$ 5. $f'(x) = 5$ 7. $f'(x) = 3x^2 - 4$

9. $1 - 3 \cos x$ 11. $f'(t) = t^3$ 13. $A'(s) = 60/s^6$

15. $g'(t) = -\frac{3}{2}t^{-7/4}$ 17. $y' = 3e^x - \frac{4}{3}x^{-4/3}$

19. $F'(x) = \frac{5}{32}x^4$ 21. $y' = \frac{3}{2}\sqrt{x} + \frac{2}{\sqrt{x}} - \frac{3}{2x\sqrt{x}}$
 23. $y' = 0$ 25. $g'(y) = -10A/y^{11} - B \sin y$
 27. $f'(x) = k(2x - a - b)$ 29. $u' = \frac{1}{5}t^{-4/5} + 10t^{3/2}$
 31. $G'(y) = -10A/y^{11} + Be^y$ 33. $y = \frac{1}{4}x + \frac{3}{4}$
 35. $y = -3\sqrt{3}x + 3 + \pi\sqrt{3}$, $y = x/(3\sqrt{3}) + 3 - \pi/(9\sqrt{3})$
 37. Tangent: $y = 2x + 2$; normal: $y = -\frac{1}{2}x + 2$
 39. $y = \frac{3}{2}x + \frac{1}{2}$



41. $45x^{14} - 15x^2$
 43. $f'(x) = 4x^3 - 9x^2 + 16$, $f''(x) = 12x^2 - 18x$
 45. $g'(t) = -2 \sin t - 3 \cos t$, $g''(t) = -2 \cos t + 3 \sin t$
 47. $\left. \frac{dL}{dA} \right|_{A=12} = 1.718$ means that a 12-year old rock bass grows at a rate of 1.718 inches/year.
 49. (a) 0.926 cm/s; 0.694 cm/s; 0
 (b) 0; -92.6 (cm/s)/cm; -185.2 (cm/s)/cm
 (c) At the center; at the edge
 51. (a) $v(t) = 3t^2 - 9t - 7$; $a(t) = 6t - 9$
 (b) $t = 4$ s (c) $t = 1.5$ s; the velocity has an absolute minimum
 53. (a) 30 mm²/mm; the rate at which the area is increasing with respect to side length as x reaches 15 mm
 (b) $\Delta A \approx 2x \Delta x$
 55. (a) (i) 5π (ii) 4.5π (iii) 4.1π
 (b) 4π (c) $\Delta A \approx 2\pi r \Delta r$
 57. $-\cos x$ 59. $(2n + 1)\pi \pm \frac{1}{3}\pi$, n an integer
 63. $y = 12x - 15$, $y = 12x + 17$ 65. $y = \frac{1}{3}x - \frac{1}{3}$
 67. $(\pm 2, 4)$ 71. $y = 2x^2 - x$
 73. $a = -\frac{1}{2}$, $b = 2$ 75. 1000

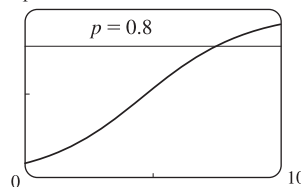
EXERCISES 3.4 | page 200

1. $1 - 2x + 6x^2 - 8x^3$ 3. $f'(x) = e^x(x^3 + 3x^2 + 2x + 2)$
 5. $g'(t) = 3t^2 \cos t - t^3 \sin t$ 7. $F'(y) = 5 + \frac{14}{y^2} + \frac{9}{y^4}$
 9. $f'(x) = \cos x - \frac{1}{2} \csc^2 x$
 11. $h'(\theta) = \csc \theta - \theta \csc \theta \cot \theta + \csc^2 \theta$
 13. $y' = (x - 2)e^x/x^3$ 15. $g'(x) = 5/(2x + 1)^2$
 17. $y' = \frac{x^2(3 - x^2)}{(1 - x^2)^2}$ 19. $y' = \frac{2t(-t^4 - 4t^2 + 7)}{(t^4 - 3t^2 + 1)^2}$
 21. $y' = (r^2 - 2)e^r$ 23. $f'(\theta) = \frac{\sec \theta \tan \theta}{(1 + \sec \theta)^2}$
 25. $y' = (x \cos x - 2 \sin x)/x^3$ 27. $y' = 2v - 1/\sqrt{v}$
 29. $f'(t) = \frac{4 + t^{1/2}}{(2 + \sqrt{t})^2}$ 31. $f'(x) = \frac{-ACe^x}{(B + Ce^x)^2}$

33. $f'(x) = \frac{2cx}{(x^2 + c)^2}$ 35. $y = \frac{1}{2}x + \frac{1}{2}$
 37. $y = 2x$; $y = -\frac{1}{2}x$ 39. $(x^4 + 4x^3)e^x$; $(x^4 + 8x^3 + 12x^2)e^x$
 41. $\theta \cos \theta + \sin \theta$; $2 \cos \theta - \theta \sin \theta$
 43. $f^{(n)}(x) = e^x(n + x)$
 49. (a) -16 (b) $-\frac{20}{9}$ (c) 20
 51. (a) 0 (b) $-\frac{2}{3}$
 53. (a) $y' = xg'(x) + g(x)$
 (b) $y' = \frac{g(x) - xg'(x)}{[g(x)]^2}$ (c) $y' = \frac{xg'(x) - g(x)}{x^2}$
 55. $\frac{dv}{d[S]} = \frac{0.0021}{(0.015 + [S])^2}$ is the rate of change of the enzymatic reaction rate with respect to the concentration of the substrate.
 57. -0.2436 K/min 59. Two, $(-2 \pm \sqrt{3}, \frac{1}{2}(1 \mp \sqrt{3}))$
 61. (c) $3e^{3x}$ 63. (b) $y' = -2x(2x^2 + 1)/(x^4 + x^2 + 1)^2$

EXERCISES 3.5 | page 212

1. $\frac{4}{3\sqrt{1+4x^2}}$ 3. $\pi \sec^2 \pi x$ 5. $\frac{e^{\sqrt{x}}}{2\sqrt{x}}$
 7. $F'(x) = 10x(x^4 + 3x^2 - 2)^4(2x^2 + 3)$
 9. $F'(x) = -\frac{1}{\sqrt{1-2x}}$ 11. $f'(z) = -\frac{2z}{(z^2 + 1)^2}$
 13. $y' = -3x^2 \sin(a^3 + x^3)$ 15. $h'(t) = 3t^2 - 3^t \ln 3$
 17. $y' = e^{-kx}(-kx + 1)$
 19. $y' = 8(2x - 5)^3(8x^2 - 5)^{-4}(-4x^2 + 30x - 5)$
 21. $y' = (\cos x - x \sin x)e^{x \cos x}$ 23. $y' = \frac{-12x(x^2 + 1)^2}{(x^2 - 1)^4}$
 25. $y' = 4 \sec^2 x \tan x$ 27. $y' = (r^2 + 1)^{-3/2}$
 29. $y' = 2 \cos(\tan 2x) \sec^2(2x)$ 31. $y' = 2^{\sin \pi x}(\pi \ln 2) \cos \pi x$
 33. $y' = -2 \cos \theta \cot(\sin \theta) \csc^2(\sin \theta)$
 35. $y' = \frac{-\pi \cos(\tan \pi x) \sec^2(\pi x) \sin \sqrt{\sin(\tan \pi x)}}{2\sqrt{\sin(\tan \pi x)}}$
 37. $y' = -2x \sin(x^2)$; $y'' = -4x^2 \cos(x^2) - 2 \sin(x^2)$
 39. $e^{\alpha x}(\beta \cos \beta x + \alpha \sin \beta x)$; $e^{\alpha x}[(\alpha^2 - \beta^2) \sin \beta x + 2\alpha\beta \cos \beta x]$
 41. $y = 20x + 1$ 43. $y = -x + \pi$
 45. 24 47. (a) 30 (b) 36
 49. (a) $F'(x) = e^x f'(e^x)$ (b) $G'(x) = e^{f(x)} f'(x)$
 51. 120 53. $-2^{50} \cos 2x$
 55. $v(t) = \frac{5}{2}\pi \cos(10\pi t)$; $a(t) = -25\pi^2 \sin(10\pi t)$
 57. $m'(t) = e^{-t} \cos t$
 59. (a) $C'(10) \approx 0.00752$ (mg/mL)/min
 (b) The BAC is decreasing at a rate of about 0.00316 (mg/mL)/min.
 61. (a) 1 (b) $\frac{dp}{dt} = \frac{kae^{-kt}}{(1 + ae^{-kt})^2}$
 (c) 1 $t \approx 7.4$ hours



63. (a) $y' = -(y + 2 + 6x)/x$
 (b) $y = (4/x) - 2 - 3x$, $y' = -(4/x^2) - 3$
65. $y' = -\frac{x^2}{y^2}$ 67. $y' = \frac{2x + y}{2y - x}$
69. $y' = \frac{3y^2 - 5x^4 - 4x^3y}{x^4 + 3y^2 - 6xy}$ 71. $y' = \tan x \tan y$
73. $y' = \frac{y(y - e^{x/y})}{y^2 - xe^{x/y}}$ 75. $y' = \frac{e^y \sin x + y \cos(xy)}{e^y \cos x - x \cos(xy)}$
77. $y = -x + 2$ 79. $y = x + \frac{1}{2}$
81. $\frac{dA}{dp} = \frac{1}{\rho q - e^{qA}}$ 83. $dV/dt = 3x^2 dx/dt$
85. $48 \text{ cm}^2/\text{s}$ 87. $80 \text{ cm}^3/\text{min}$
89. (a) $\frac{dm}{dt} = \pi r^2 L(1 - k^2) \frac{dp}{dt}$
 (b) $\frac{dm}{dt} = -2\pi r^2 L(\rho - 1)k \frac{dk}{dt}$
91. $\frac{dr}{da} = \frac{2}{\sqrt{1 + 8s}} \frac{ds}{da}$ 93. $\frac{dP}{dx} = -\frac{8\eta lv}{R^3} R'(x)$

EXERCISES 3.6 | page 220

1. About 235
3. (a) $100(4.2)^t$ (b) ≈ 7409 (c) $\approx 10,632$ bacteria/h
 (d) $(\ln 100)/(\ln 4.2) \approx 3.2$ h
5. (a) 1508 million, 1871 million (b) 2161 million
 (c) 3972 million; wars in the first half of century, increased life expectancy in second half
7. (a) $100 \times 2^{-t/30}$ mg (b) ≈ 9.92 mg
 (c) ≈ -0.229 mg/year (d) ≈ 199.3 years
9. (a) $-\frac{\ln 2}{\ln 0.945} \approx 12.25$ years (b) $-\frac{\ln 5}{\ln 0.945} \approx 28.45$ years
11. $\approx 57,104$ years
13. (a) $\approx 137^\circ\text{F}$ (b) ≈ 116 min
15. (a) 13.3°C (b) ≈ 67.74 min
17. (a) ≈ 64.5 kPa (b) ≈ 39.9 kPa

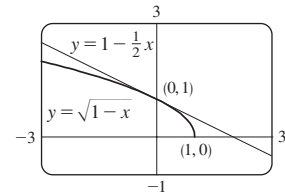
EXERCISES 3.7 | page 229

1. The differentiation formula is simplest.
3. $f'(x) = \frac{\cos(\ln x)}{x}$ 5. $f'(x) = \frac{3}{(3x - 1)\ln 2}$
7. $f'(x) = \frac{1}{5x\sqrt[3]{(\ln x)^4}}$ 9. $f'(x) = \frac{\sin x}{x} + \cos x \ln(5x)$
11. $F'(t) = \frac{6}{2t + 1} - \frac{12}{3t - 1}$ 13. $g'(x) = \frac{2x^2 - 1}{x(x^2 - 1)}$
15. $y' = \frac{10x + 1}{5x^2 + x - 2}$ 17. $y' = \frac{-x}{1 + x}$
19. $y' = \frac{1}{\ln 10} + \log_{10} x$
21. $y' = x + 2x \ln(2x)$; $y'' = 3 + 2 \ln(2x)$
23. $f'(x) = \frac{2x - 1 - (x - 1) \ln(x - 1)}{(x - 1)[1 - \ln(x - 1)]^2}$; $(1, 1 + e) \cup (1 + e, \infty)$
25. $y = 3x - 9$ 27. 1

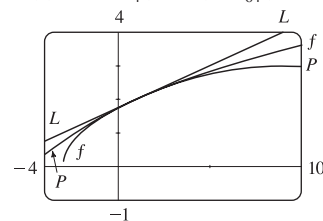
29. $\frac{dt}{dc} = \frac{3 + (9c - 4)/\sqrt{9c^2 - 8c}}{3c + \sqrt{9c^2 - 8c}}$ is the rate of change of dialysis duration as the initial urea concentration increases.
31. $\frac{da}{dN} = -\frac{5370}{(\ln 2)N}$ is the rate of change of the estimated age with respect to an increase in the measured amount of ^{14}C .
33. $y' = (2x + 1)^5(x^4 - 3)^6 \left(\frac{10}{2x + 1} + \frac{24x^3}{x^4 - 3} \right)$
35. $y' = \frac{\sin^2 x \tan^4 x}{(x^2 + 1)^2} \left(2 \cot x + \frac{4 \sec^2 x}{\tan x} - \frac{4x}{x^2 + 1} \right)$
37. $y' = x^x(\ln x + 1)$ 39. $y' = (\cos x)^x(-x \tan x + \ln \cos x)$
41. $y' = (\tan x)^{1/x} \left(\frac{\sec^2 x}{x \tan x} - \frac{\ln \tan x}{x^2} \right)$ 43. $y' = \frac{2 \tan^{-1} x}{1 + x^2}$
45. $y' = -\frac{\sin \theta}{1 + \cos^2 \theta}$ 47. $y' = \frac{1}{2(1 + x^2)}$ 49. $\pi/2$
51. $y' = \frac{2x}{x^2 + y^2 - 2y}$ 53. $f^{(n)}(x) = \frac{(-1)^{n-1}(n-1)!}{(x-1)^n}$

EXERCISES 3.8 | page 237

1. $L(x) = -10x - 6$ 3. $L(x) = -x + \pi/2$
5. $\sqrt{1-x} \approx 1 - \frac{1}{2}x$;
 $\sqrt{0.9} \approx 0.95$,
 $\sqrt{0.99} \approx 0.995$



7. $-1.204 < x < 0.706$ 9. $-0.045 < x < 0.055$
11. 32.08 15. $L(s) = p + p(1-p)s$
17. A 5% increase in the radius corresponds to a 20% increase in blood flow.
19. (a) $x_2 \approx 2.3$, $x_3 \approx 3$ (b) No
21. 1.1797 23. -0.724492 , 1.220744
25. 1.412391, 3.057104
27. -1.93822883 , -1.21997997 , 1.13929375 , 2.98984102
29. -1.97806681 , -0.82646233
31. 0.21916368, 1.08422462
35. $T_3(x) = 1 + x + \frac{1}{2}x^2 + \frac{1}{6}x^3$
37. $T_3(x) = 1 - (x-1) + (x-1)^2 - (x-1)^3 + (x-1)^4$
39. $L(x) = \frac{1}{4}x + \frac{7}{4}$
 $P(x) \approx 2 + \frac{1}{4}(x-1) - \frac{1}{64}(x-1)^2$



$P(x)$ is a better approximation than $L(x)$.